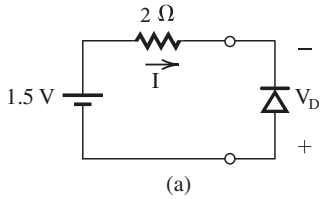
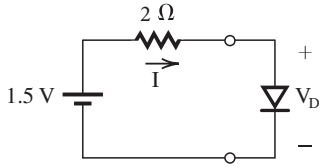
4.1



(a)



(b)

(a) The diode is reverse biased, thus

$I = 0$ A

$V_D = -1.5$ V

(b) The diode is forward biased, thus

$V_D = 0$ V

$$I = \frac{1.5 \text{ V}}{2\ \Omega} = 0.75 \text{ A}$$

4.2 Refer to Fig. P4.2.

(a) Diode is conducting, thus

$V = -5$ V

$$I = \frac{+5-(-5)}{10\ \text{k}\Omega} = 1.0 \text{ mA}$$

(b) Diode is reverse biased, thus

$I = 0$

$V = +5$ V

(c) Diode is conducting, thus

$V = +5$ V

$$I = \frac{+5-(-5)}{10\ \text{k}\Omega} = 1.0 \text{ mA}$$

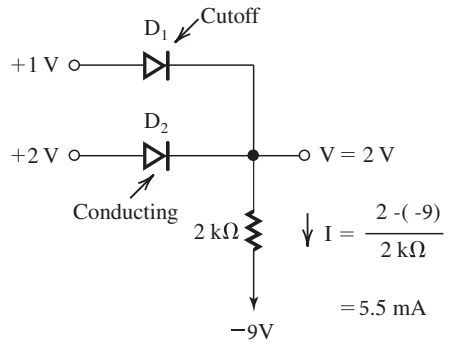
(d) Diode is reverse biased, thus

$I = 0$

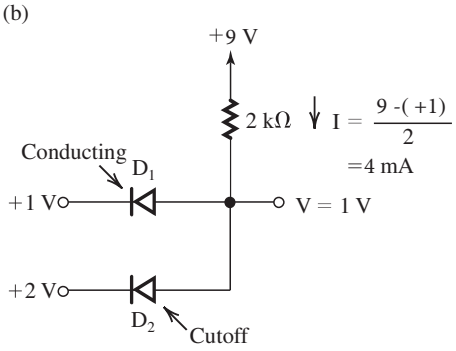
$V = -5$ V

4.3

(a)

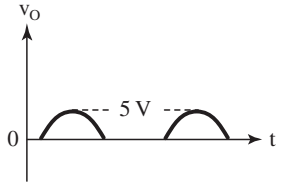


(b)



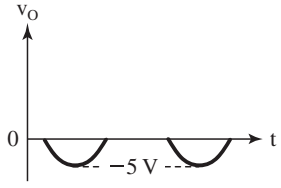
4.4

(a)



$V_{p^+} = 5$ V $\quad V_{p^-} = 0$ V $\quad f = 1$ kHz

(b)



$V_{p^+} = 0$ V $\quad V_{p^-} = -5$ V $\quad f = 1$ kHz

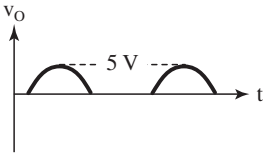
(c)



$v_O = 0$ V

Neither D_1 nor D_2 conducts, so there is no output.

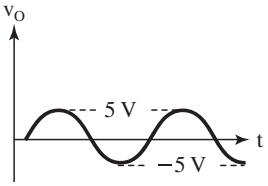
(d)



$V_{p^+} = 5$ V, $V_{p^-} = 0$ V, f = 1 kHz

Both D_1 and D_2 conduct when $v_I > 0$

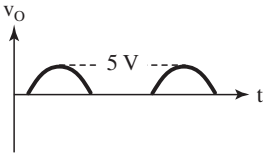
(e)



$V_{p^+} = 5$ V, $V_{p^-} = -5$ V, f = 1 kHz

D_1 conducts when $v_I > 0$ and D_2 conducts when $v_I < 0$. Thus the output follows the input.

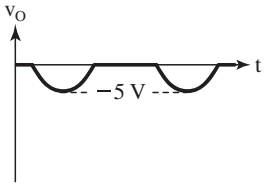
(f)



$V_{p^+} = 5$ V, $V_{p^-} = 0$ V, f = 1 kHz

D_1 is cut off when $v_I < 0$

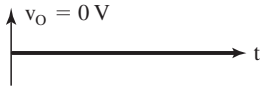
(g)



$V_{p^+} = 0$ V, $V_{p^-} = -5$ V, f = 1 kHz

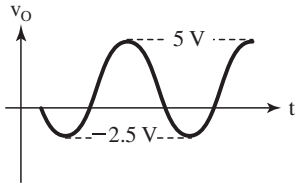
D_1 shorts to ground when $v_I > 0$ and is cut off when $v_I < 0$ whereby the output follows v_I.

(h)



$v_O = 0$ V ~ The output is always shorted to ground as D_1 conducts when $v_I > 0$ and D_2 conducts when $v_I < 0$.

(i)



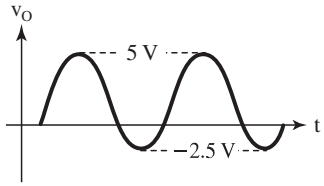
$V_{p^+} = 5$ V, $V_{p^-} = -2.5$ V, f = 1 kHz

When $v_I > 0$, D_1 is cut off and v_O follows v_I.

When $v_I < 0$, D_1 is conducting and the circuit becomes a voltage divider where the negative peak is

$$\frac{1\text{ k}\Omega}{1\text{ k}\Omega + 1\text{ k}\Omega} \times -5\text{ V} = -2.5\text{ V}$$

(j)

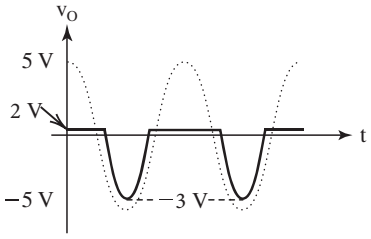


$V_{p^+} = 5$ V, $V_{p^-} = -2.5$ V, f = 1 kHz

When $v_I > 0$, the output follows the input as D_1 is conducting.

When $v_I < 0$, D_1 is cut off and the circuit becomes a voltage divider.

(k)



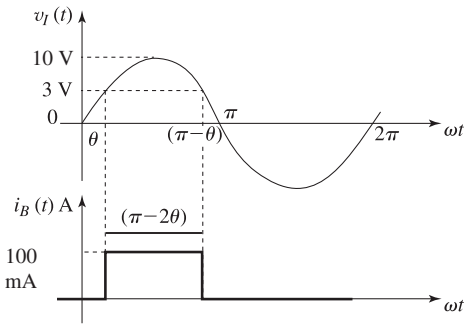
$V_{p+} = 2$ V, $V_{p^-} = -3$ V, f = 1 k Hz

When $v_I > 0$, D_1 is cut off and D_2 is conducting. The output becomes 2 V.

When $v_I < 0$, D_1 is conducting and D_2 is cut off.

The output becomes: $v_{O=}v_I + 2$ V

4.5



From Fig. P4.5 we see that when $v_I < V_B$; that is, $v_I < 3$ V, D_1 will be conducting the current I and i_B will be zero. When v_I exceeds the battery voltage (3 V), D_1 cuts off and D_2 conducts, thus steering I into the battery. Thus, i_B will have the waveform shown in the figure. Its peak value will be 60 mA. To obtain the average value, we first determine the conduction angle of D_2, $(\pi - 2\theta)$, where

$$\theta = \sin^{-1}\left(\frac{3}{6}\right) = 30^\circ$$

Thus

$$\pi - 2\theta = 180^\circ - 60 = 120^\circ$$

The average value of i_B will be

$$i_B|_{av} = \frac{60 \times 120^\circ}{360^\circ} = 20 \text{ mA}$$

If the peak value of v_I is reduced by 10%, i.e. from 6 V to 5.4 V, the peak value of i_B does not change. The conduction angle of D_2, however, changes since θ now becomes

$$\theta = \sin^{-1}\left(\frac{3}{5.4}\right) = 33.75^\circ$$

and thus

$$\pi - 2\theta = 112.5^\circ$$

Thus the average value of i_B becomes

$$i_B|_{av} = \frac{100 \times 112.5^\circ}{360^\circ} = 31.25 \text{ mA}$$

4.6

A	B	X	Y
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

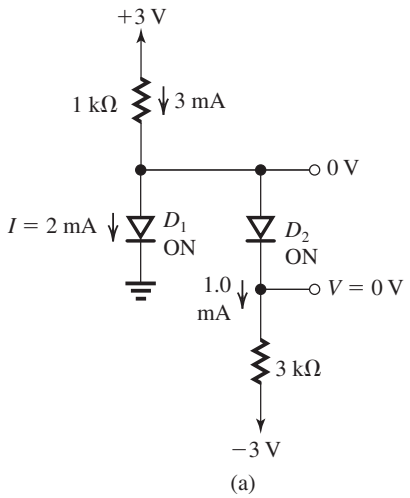
X = AB, Y = A + B

X and Y are the same for

A = B

X and Y are opposite if A ≠ B

4.7

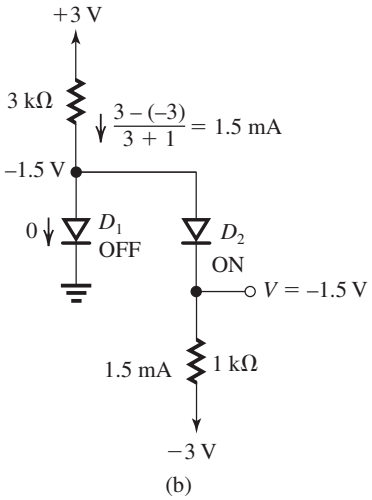


(a)

(a) If we assume that both D_1 and D_2 are conducting, then $V = 0$ V and the current in D_2 will be $[0 - (-3)]/3 = 1$ mA. The current in the 1 kΩ will be $(3 - 0)/1 = 3$ mA. A node equation at the common anodes node yields a 2 mA current in D_1. Thus, we obtain the results shown in Fig. (a):

$I = 2$ mA

$V = 0$ V



(b)

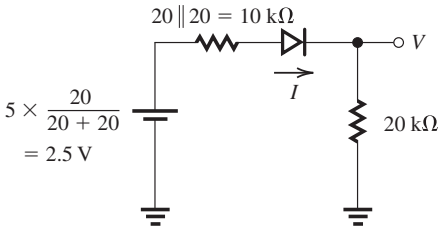
(b) In (b), the two resistors are interchanged. With some reasoning, we can see that with D_2

conducting, the intermediate node V will be below 0 V due to the voltage division between the two resistors. Thus, D_1 is off giving the results shown in Fig. (b):

$I = 0$ mA

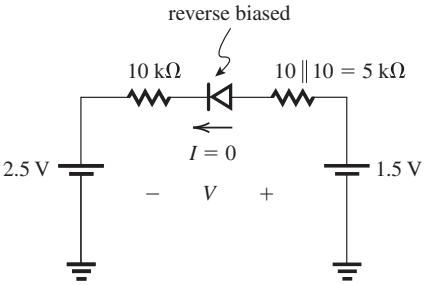
$V = -1.5$ V

4.8 The analysis is shown on the circuit diagrams below.



$I = \frac{2.5}{10+20} = 0.083$ mA $\quad V = 0.083 \times 20 = 1.67$ V

(a)



$V = 1.5 - 2.5 = -1$ V

(b)

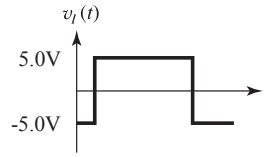
4.9 $R \geq \frac{120\sqrt{2}}{40} \geq 4.2$ kΩ

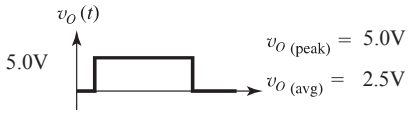
The largest reverse voltage appearing across the diode is equal to the peak input voltage:

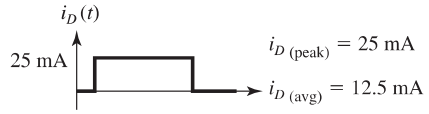
$120\sqrt{2} = 169.7$ V

4.10 For $v_I > 0$ V : D is on, $v_O = v_I$, $i_D = v_I/R$

For $v_I < 0$ V: D is off , $v_O = 0$,$i_D = 0$







reverse diode voltage: $V_R = v_O = v_I$

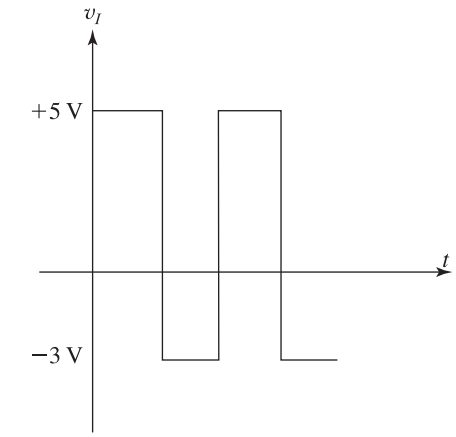
$V_{R\,(max)} = 5.0$V

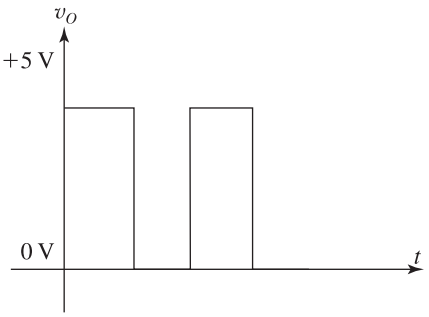
4.11

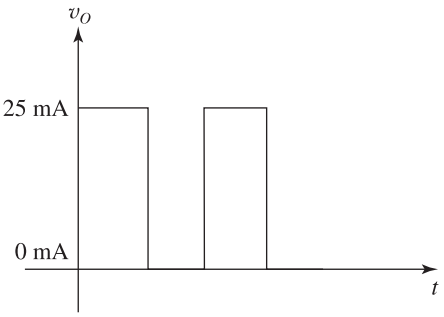
Using an ideal diode model,

$i_{D(peak)} = \frac{v_{O(peak)}}{R} = \frac{5}{0.2} = 25$ mA

$i_{D(avg)} = \frac{i_{D(peak)}}{2} = 12.5$ mA





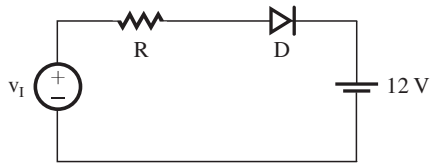


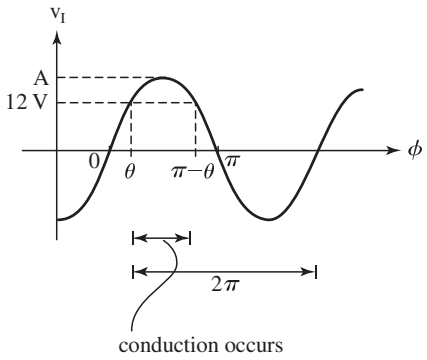
4.12 The peak current is given by

$\frac{24-12}{R} = 50$

$\Rightarrow R = \frac{24-12}{50} = 0.24$ kΩ $= 240\Omega$

4.13





$v_I = A \sin\theta = 12 \sim$ conduction through D occurs

For a conduction angle $(\pi - 2\theta)$ that is 25% of a cycle

$$\frac{\pi - 2\theta}{2\pi} = \frac{1}{4}$$

$$\theta = \frac{\pi}{4}$$

$A = 12/\sin\theta = 17$ V

$\therefore$ Peak-to-peak sine wave voltage

$= 2A = 34$ V

Given the average diode current to be

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{A\sin\phi - 12}{R}\, d\phi = 100 \text{ mA}$$

$$\frac{1}{2\pi}\left(\frac{-17\cos\phi - 12\phi}{R}\right)_{\phi=0.25\pi}^{\phi=0.75\pi} = 0.1$$

$R = 8.3\ \Omega$

$$\text{Peak diode current} = \frac{A - 12}{R} = 0.6 \text{ A}$$

Peak reverse voltage $= A + 12 = 29$ V

For resistors specified to only one significant digit and peak-to-peak voltage to the nearest volt, choose $A = 17$ so the peak-to-peak sine wave voltage $= 34$ V and $R = 8\ \Omega$.

Conduction starts at $v_I = A\sin\theta = 12$

$17\sin\theta = 12$

$$\theta = \left(\frac{\pi}{4}\right) \text{ rad}$$

Conduction stops at $\pi - \theta$.

$\therefore$ Fraction of cycle that current "ows is

$$\frac{\pi - 2\theta}{2\pi} \times 100 = 25\%$$

Average diode current $=$

$$\frac{1}{2\pi}\left(\frac{-17\cos\phi - 12\phi}{8}\right)_{\phi=\pi/4}^{\phi=3\pi/4} = 103 \text{ mA}$$

Peak diode current

$$= \frac{17 - 12}{8} = 0.625 \text{ A}$$

Peak reverse voltage $=$

$A + 12 = 29$ V

4.14

V	RED	GREEN	
3 V	ON	OFF	- D_1 conducts
0	OFF	OFF	- No current "ows
–3 V	OFF	ON	- D_2 conducts

4.15 The maximum reverse voltage arises when

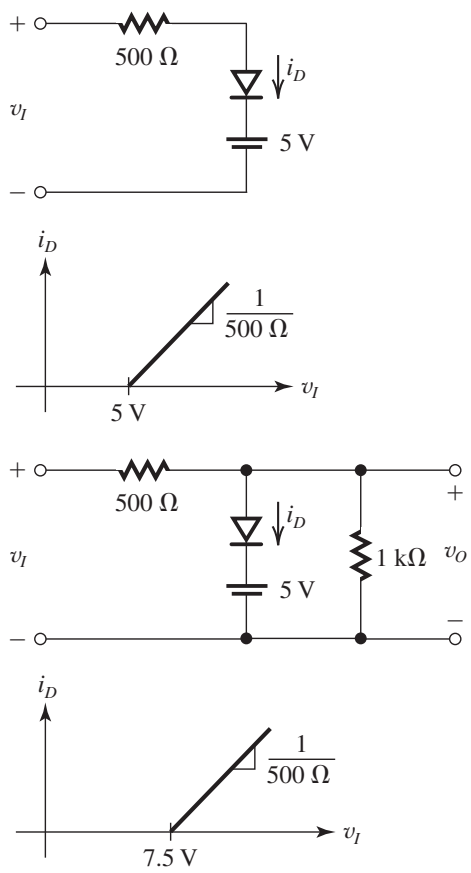
$v_I = V_+ - 10 = 3 - 10 = -7$ V

The maximum forward current arises when

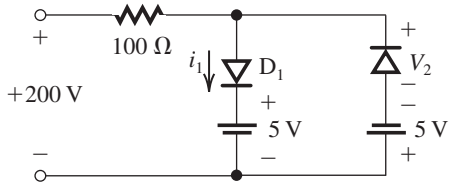
$v_I = I_{max} \times R + V_+ = 50 \times 0.1 + 3 = 8$ V

Thus, the safe operating range is
$-7 \text{ V} \le v_I \le 8 \text{ V}$.

4.16



4.17



Assuming ideal diodes with $v_I = +200$ V, the diode D_1 is conducting with zero voltage drop across it. The peak current is thus

$$i_1 = \frac{200 - 5}{100} = 1.95 \text{ A}$$

and the peak voltage is

$$v_2 = 5 - (-5) = 10 \text{ V}$$

4.18 $V_T = \dfrac{kT}{q}$

where

$k = 1.38 \times 10^{-23}$ J/K $= 8.62 \times 10^{-5}$eV/K

$T = 273 + x°C$

$q = 1.60 \times 10^{-19}$C

Thus

$V_T = 8.62 \times 10^{-5} + (273 \times x°C)$, V

x [°C]	V_T [mV]
−55	18.8
0	23.5
+40	27.0
+125	34.3

for $V_T = 25$ mV at 17°C

4.19 $i = I_S e^{v/0.025}$

$\therefore 100{,}000{,}000 I_S = I_S e^{v/0.025}$

$v = 0.461$ V

At $v = 0.7$ V,

$i = I_S e^{0.7/0.025} = 1.45 \times 10^{12} I_S$

4.20 $I_1 = I_S e^{0.7/V_T} = 10^{-3}$

$i_2 = I_S e^{0.6/V_T}$

$$\frac{i_2}{i_1} = \frac{i_2}{10^{-3}} = e^{\frac{0.6 - 0.7}{0.025}}$$

$i_2 = 18.32\ \mu$A

4.21 $I = I_S e^{V_D/V_T}$

$10^{-3} = I_S e^{0.7/V_T}$ (1)

For $V_D = 0.71$ V,

$I = I_S e^{0.71/V_T}$ (2)

Combining (1) and (2) gives

$I = 10^{-3} e^{(0.71-0.7)/0.025}$

$= 1.49$ mA

For $V_D = 0.8$ V,

$I = I_S e^{0.8/V_T}$ (3)

Combining (1) and (3) gives

$I = 10^{-3} \times e^{(0.8-0.7)/0.025}$

$= 54.6$ mA

Similarly, for $V_D = 0.69$ V we obtain

$I = 10^{-3} \times e^{(0.69-0.7)/0.025}$

$= 0.67$ mA

and for $V_D = 0.6$ V we have

$I = 10^{-3} e^{(0.6-0.7)/0.025}$

$= 18.3\ \mu$A

To increase the current by a factor of 2, V_D must be increased by $\triangle V_D$,

$2 = e^{\triangle V_D/0.025}$

$\Rightarrow \triangle V_D = 0.025 \ln 2 = 17.3$ mV

4.22 I_S can be found by using $I_S = I_D \cdot e^{-V_D/V_T}$.

Let a decrease by a factor of 10 in I_D result in a decrease of V_D by ΔV:

$I_D = I_S e^{V_D/V_T}$

$\frac{I_D}{10} = I_S e^{(V_D-\Delta V)/V_T} = I_S e^{V_D/V_T} \cdot e^{-\Delta V/V_T}$

Taking the ratio of the above two equations, we have

$10 = e^{\Delta V/V_T} \Rightarrow \Delta V \simeq 60$ mV

Thus the result in each case is a decrease in the diode voltage by 60 mV.

(a) $V_D = 0.750$ V, $I_D = 1$ A
$\Rightarrow I_S = 9.36 \times 10^{-14}$ A;

10% of I_D gives $V_D = 0.69$ V

(b) $V_D = 0.650$ V, $I_D = 1$ m A
$\Rightarrow I_S = 5.11 \times 10^{-15}$ A;

10% of I_D gives $V_D = 0.59$ V

(c) $V_D = 0.650$ V, $I_D = 10\ \mu$A
$\Rightarrow I_S = 5.11 \times 10^{-17}$ A;

10% of I_D gives $V_D = 0.59$ V

(d) $V_D = 0.700$ V, $I_D = 10$ mA
$\Rightarrow I_S = 6.91 \times 10^{-15}$ A;

10% of I_D gives $V_D = 0.64$ V

4.23 $I_D = I_S\ e^{V_D/V_T}$

(a) $V_D = 0.70$ V, $I_D = 10$ mA
$\Rightarrow I_S = 6.91 \times 10^{-15}$ A;

When $V_D = 0.75$ V

I_D=73.8mA

(b) $V_D = 0.70$ V, $I_D = 1$ m A
$\Rightarrow I_S = 6.91 \times 10^{-16}$ A;

When $V_D = 0.75$ V

I_D=7.38mA

(c) $V_D = 0.80$ V, $I_D = 10$ A
$\Rightarrow I_S = 1.27 \times 10^{-13}$ A;

When $V_D = 0.75$ V

I_D=1.36A

(d) $V_D = 0.70$ V, $I_D = 0.1$ m A
$\Rightarrow I_S = 6.91 \times 10^{-17}$ A;

When $V_D = 0.75$ V

I_D=0.738mA

(e) $V_D = 0.6$ V, $I_D = 10\ \mu$A
$\Rightarrow I_S = 3.78 \times 10^{-16}$ A

When $V_D = 0.75$ V

I_D= 4.04mA

4.24 The voltage across three diodes in series is 2.0 V; thus the voltage across each diode must be 0.667 V. Using $I_D = I_S e^{V_D/V_T}$, the required current I is found to be 3.9 mA.

If 1 mA is drawn away from the circuit, I_D will be 2.9 mA, which would give $V_D = 0.6548$ V, giving an output voltage of 1.98 V. The change in output voltage is -22 mV.

If temperature increase of 40°C.Is increase to 8Is, U_D will decrease

4.25 Connecting an identical diode in parallel would reduce the current in each diode by a factor of 2. Writing expressions for the currents, we have

$I_D = I_S e^{V_D/V_T}$

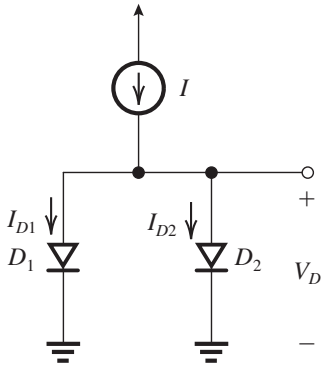
$\frac{I_D}{2} = I_S e^{(V_D-\Delta V)/V_T} = I_S e^{V_D/V_T} \cdot e^{-\Delta V/V_T}$

Taking the ratio of the above two equations, we have

$2 = e^{\Delta V/V_T} \Rightarrow \Delta V = 17.3$ mV

Thus the result is a decrease in the diode voltage by 17.3 mV.

4.26



$$I_{D1} = I_{S1}e^{V_D/V_T} \quad (1)$$

$$I_{D2} = I_{S2}e^{V_D/V_T} \quad (2)$$

Summing (1) and (2) gives

$$I_{D1} + I_{D2} = (I_{S1} + I_{S2})e^{V_D/V_T}$$

But

$$I_{D1} + I_{D2} = I$$

Thus

$$I = (I_{S1} + I_{S2})\, e^{V_D/V_T} \quad (3)$$

From Eq. (3) we obtain

$$V_D = V_T \ln\left(\frac{I}{I_{S1} + I_{S2}}\right)$$

Also, Eq. (3) can be written as

$$I = I_{S1}\, e^{V_D/V_T}\left(1 + \frac{I_{S2}}{I_{S1}}\right) \quad (4)$$

Now using (1) and (4) gives

$$I_{D1} = \frac{I}{1 + (I_{S2}/I_{S1})} = I\frac{I_{S1}}{I_{S1} + I_{S2}}$$

We similarly obtain

$$I_{D2} = \frac{I}{1 + (I_{S1}/I_{S2})} = I\frac{I_{S2}}{I_{S1} + I_{S2}}$$

4.27 We can write a node equation at the anodes:

$$I_{D2} = I_1 - I_2 = 5 \text{ mA}$$

$$I_{D1} = I_2 = 3 \text{ mA}$$

We can write the following equation for the diode voltages:

$$V = V_{D2} - V_{D1}$$

If D_2 has saturation current I_S, then D_1, which is 20 times larger, has saturation current $20I_S$. Thus we can write

$$I_{D2} = I_S e^{V_{D2}/V_T}$$

$$I_{D1} = 20I_S e^{V_{D1}/V_T}$$

Taking the ratio of the two equations above, we have

$$\frac{I_{D2}}{I_{D1}} = \frac{5}{3} = \frac{1}{20}e^{(V_{D2}-V_{D1})/V_T} = \frac{1}{20}e^{V/V_T}$$

$$\Rightarrow V = 0.025 \ln\left(\frac{100}{3}\right) = 87.7 \text{ mV}$$

To instead achieve $V = 60$ mV, we need

$$\frac{I_{D2}}{I_{D1}} = \frac{I_1 - I_2}{I_2} = \frac{1}{20}e^{0.06/0.025} = 0.55$$

Solving the above equation with I_1 still at 8 mA, we find $I_2 = 5.16$ mA.

4.28 We can write the following node equation at the diode anodes:

$$I_{D2} = 1\text{mA} - V/R$$

$$I_{D1} = V/R$$

We can write the following equation for the diode voltages:

$$V = V_{D2} - V_{D1}$$

We can write the following diode equations:

$$I_{D2} = I_S e^{V_{D2}/V_T}$$

$$I_{D1} = I_S e^{V_{D1}/V_T}$$

Taking the ratio of the two equations above, we have

$$\frac{I_{D2}}{I_{D1}} = \frac{1\text{mA} - V/R}{V/R} = e^{(V_{D2}-V_{D1})/V_T} = e^{V/V_T}$$

To achieve $V = 50$ mV, we need

$$\frac{I_{D2}}{I_{D1}} = \frac{1\text{mA} - 0.05/R}{0.05/R} = e^{0.05/0.025} = 7.39$$

Solving the above equation, we have

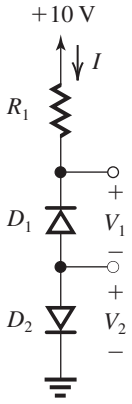
$$R = 419.5\ \Omega$$

4.29 For a diode conducting a constant current, the diode voltage decreases by approximately 2 mV per increase of 1°C.

$T = -20$°C corresponds to a temperature decrease of 40°C, which results in an increase of the diode voltage by 80 mV. Thus $V_D = 770$ mV.

$T = +85$°C corresponds to a temperature increase of 65°C, which results in a decrease of the diode voltage by 130 mV. Thus $V_D = 560$ mV.

4.30



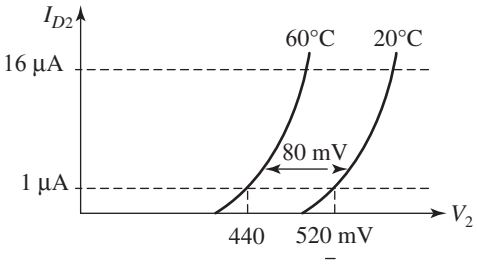
At 20°C:

$V_{R1} = V_2 = 520\ \text{mV}$

$R_1 = 520\ \text{k}\Omega$

$I = \dfrac{520\ \text{mV}}{520\ \text{k}\Omega} = 1\ \mu\text{A}$

Since the reverse current doubles for every 10°C rise in temperature, at 60°C, $I = 16\ \mu\text{A}$



$V_2 = 440 + 2.3 \times 1 \times 25 \log 16$

$= 509.2\ \text{mV}$

$V_{R1} = 16\mu\text{A} \times 520\ \text{k}\Omega = 8.32\ \text{V}$

At 0°C, $I = \dfrac{1}{4}\ \mu\text{A}$

$V_2 = 560 - 2.3 \times 1 \times 25 \log 4$

$= 525.4\ \text{mV}$

$V_{R1} = \dfrac{1}{4} \times 520 = 0.13\ \text{V}$

4.31 For a diode conducting a constant current, the diode voltage decreases by approximately 2 mV per increase of 1°C.

A decrease in V_D by 100 mV corresponds to a junction temperature increase of 50°C.

The power dissipation is given by

$PD = (10\ \text{A})\ (0.6\ \text{V}) = 6\ \text{W}$

The thermal resistance is given by

$\dfrac{\Delta T}{P_D} = \dfrac{50°\text{C}}{6\ \text{W}} = 8.33°\text{C/W}$

4.32 Given two different voltage/current measurements for a diode, we can write

$I_{D1} = I_S e^{V_{D1}/V_T}$

$I_{D2} = I_S e^{V_{D2}/V_T}$

Taking the ratio of the above two equations, we have

$\dfrac{I_{D1}}{I_{D2}} = e^{(V_{D1}-V_{D2})/V_T} \Rightarrow V_{D1} - V_{D2}$

$= V_T \ln\left(\dfrac{I_{D1}}{I_{D2}}\right)$

For $I_D = 1$ mA, we have

$\Delta V = V_T \ln\left(\dfrac{1 \times 10^{-3}\ \text{A}}{10\ \text{A}}\right) = -230\ \text{mV}$

$\Rightarrow V_D = 570\ \text{mV}$

For $I_D = 3$ mA, we have

$\Delta V = V_T \ln\left(\dfrac{3 \times 10^{-3}\ \text{A}}{10\ \text{A}}\right) = -202\ \text{mV}$

$\Rightarrow V_D = 598\ \text{mV}$

Assuming V_D changes by –2 mV per 1°C increase in temperature, we have, for ±20°C changes:

For $I_D = 1$ mA, $530\ \text{mV} \le V_D \le 610\ \text{mV}$

For $I_D = 3$ mA, $558\ \text{mV} \le V_D \le 638\ \text{mV}$

Thus the overall range of V_D is between 530 mV and 638 mV.

4.33 Given two different voltage/current measurements for a diode, we have

$\dfrac{I_{D1}}{I_{D2}} = e^{(V_{D1}\ V_{D2})/\ V_T} \quad \Rightarrow V_{D1} - V_{D2}$

$= V_T \ln\left(\dfrac{I_{D1}}{I_{D2}}\right)$

For the first diode, with $I_D = 0.1$ mA and

$V_D = 700$ mV, we have

$I_D = 1$ mA:

$\Delta V = V_T \ln\left(\dfrac{1.0}{0.1}\right) = 57.6\ \text{mV}$

$\Rightarrow V_D = 757.6\ \text{mV}$

$I_D = 3$ mA:

$\Delta V = V_T \ln\left(\dfrac{3}{0.1}\right) = 85\ \text{mV} \Rightarrow V_D = 785\ \text{mV}$

For the second diode, with

$I_D = 1$ A and $V_D = 700$ mV, we have

$I_D = 1.0$ mA:

$\Delta V = V_T \ln\left(\dfrac{0.001}{1}\right) = -173\ \text{mV}$

$\Rightarrow V_D = 527\ \text{mV}$

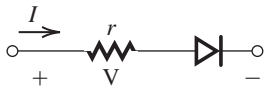
$I_D = 3$ mA:

$$\Delta V = V_T \ln\left(\frac{0.003}{1}\right) = -145 \text{ mV}$$

$$\Rightarrow V_D = 555 \text{ mV}$$

For both $I_D = 1.0$ mA and $I_D = 3$ mA, the difference between the two diode voltages is approximately 230 mV. Since, for a fixed diode current, the diode voltage changes with temperature at a constant rate (–2 mV per °C temp. increase), this voltage difference will be independent of temperature!

4.34



$$V = Ir + 2.3V_T \log\left(\frac{I}{I_S}\right)$$

Thus, a change in current, from I_1 to I_2, results in a change in voltage,

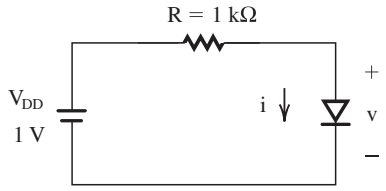
$$V_2 - V_1 = r(I_2 - I_1) + 2.3V_T \log\frac{I_2}{I_1}$$

In this case, $r = 10\Omega$, $I_1 = 1$ mA and $I_2 = 2$ mA. Thus,

$$V_2 - V_1 = 10 \times (2-1) + 2.3V_T \log\left(\frac{2}{1}\right)$$

$$= 27.3 \text{ mV}$$

4.35



$I_S = 10^{-15}\text{A} = 10^{-12}$ mA

Calculate some points

$v = 0.6$ V, $\quad i = I_S e^{v/V_T}$

$= 10^{-12} e^{0.6/0.025}$

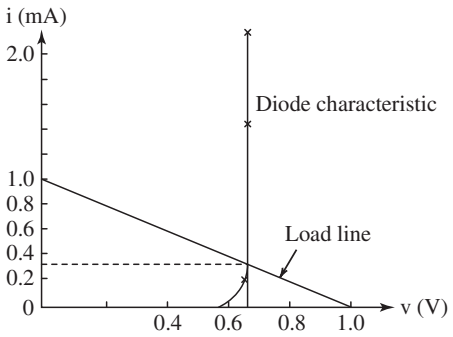
$\simeq 0.03$ mA

$v = 0.65$ V, $\quad i \simeq 0.2$ mA

$v = 0.7$ V, $\quad i \simeq 1.45$ mA

Make a sketch showing these points and load line and determine the operating point. The points for the load line are obtained using

$$I_D = \frac{V_{DD} - V_D}{R}$$



From this sketch one can see that the operating point must lie between $v = 0.65$ V to $v = 0.7$ V

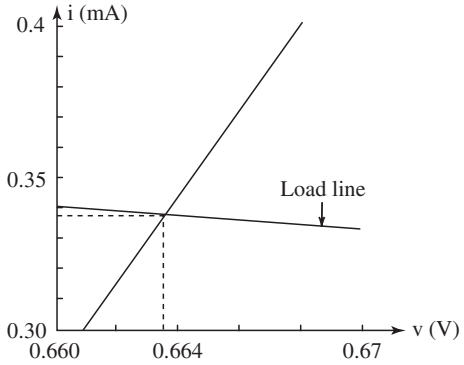
For $i = 0.3$ mA, $v = V_T \ln\left(\frac{i}{I_S}\right)$

$$= 0.025 \times \ln\left(\frac{3}{10^{-12}}\right)$$

$= 0.661$ V

For $i = 0.4$ mA, $v = 0.668$ V

Now we can refine the diagram to obtain a better estimate



From this graph we get the operating point

$i = 0.338$ mA, $v = 0.6635$ V

Now we compare graphical results with the exponential model.

At $i = 0.338$ mA

$$v = V_T \ln\left(\frac{i}{I_S}\right) = 0.025 \times \ln\left(\frac{0.338}{10^{-12}}\right)$$

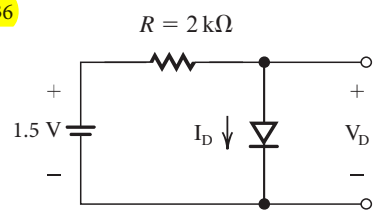
$= 0.6637$ V

The difference between the exponential model and graphical results is $= 0.6637 - 0.6635$

$= 0.0002$ V

$= 0.2$ mV

4.36



$I_S = 10^{-15}\text{A} = 10^{-12}$ mA

Use the iterative analysis procedure:

1. $V_D = 0.7$ V, $I_D = \dfrac{1.5\text{-}0.7}{2\text{ K}} = 0.4$ mA

2. $V_D = V_T \ln\left(\dfrac{I_D}{I_S}\right) = 0.025 \ln\left(\dfrac{0.4}{10^{-12}}\right)$

$= 0.6679$ V

$I_D = \dfrac{1.5\text{-}0.6679}{2\text{ K}} = 0.4161$ mA

3. $V_D = 0.025 \ln\left(\dfrac{0.4161}{10^{-12}}\right) = 0.6689$ V

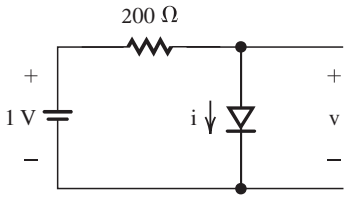
$I_D = \dfrac{1.5\text{-}0.6689}{2\text{ K}} = 0.4156$ mA

4. $V_D = 0.025 \ln\left(\dfrac{0.4156}{10^{-12}}\right) = 0.6688$ V

$I_D = \dfrac{1.5\text{-}0.6688}{2\text{ K}} = 0.4156$ mA

Stop here as we are getting almost same value of I_D and V_D

4.37



a) $I_D = \dfrac{1 - 0.7}{0.2\text{ k}\Omega} = 1.5$ mA

b) Diode has 0.7 V drop at 1 mA current. Use Eq. (4.5):

$v_2 = v_1 + 2.3 V_T \log\left(\dfrac{i_2}{i_1}\right)$

1. $v = 0.7$ V

$i = \dfrac{1 - 0.7}{0.2\text{ k}\Omega} = 1.5$ mA

2. $v = 0.7 + 2.3 \times 0.025 \log\left(\dfrac{1.5}{1}\right)$

$= 0.7101$ V

$i = \dfrac{1\text{- }0.7101}{0.2} = 1.4495$ mA

3. $v = 0.7 + 2.3 \times 0.025 \log\left(\dfrac{1.4495}{1}\right)$

$= 0.7093$ V

$i = \dfrac{1\text{ -}0.7093}{0.2\text{ k}\Omega} = 1.4535$ mA

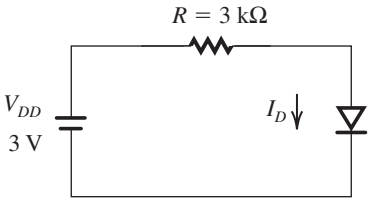
4. $v = 0.7 + 2.3 \times 0.025 \log\left(\dfrac{1.4535}{1}\right)$

$= 0.7093$ V

$i = \dfrac{1 - 0.7093}{0.2\text{ k}\Omega} = 1.4535$ mA

Stop as we are getting the same result.

4.38



a) $I_D = \dfrac{3 - 0.7}{3} = 0.767$ mA

b) $I_D = I_S e^{V_D/V_T} \Rightarrow I_S = I_D e^{-V_D/V_T}$

$I_S = 0.767 \times e^{-700/25} = 5.3 \times 10^{-16}$ A

c) I_S increases by a factor of 100

$\Rightarrow \Delta V_D = V_T \left[\ln\left(\dfrac{I_D}{100 I_S}\right) - \ln\left(\dfrac{I_D}{I_S}\right)\right]$

$= -V_T \ln 100 = -115$ mV

$\Rightarrow I_D = \dfrac{3 - (0.7 - 0.115)}{3} = \dfrac{3 - 0.585}{3}$

$= 0.805$ mA

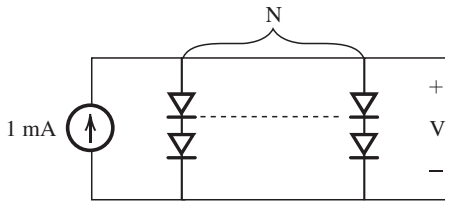
We can iterate the anlysis to improve the accuracy. With $I_D = 0.805$ mA,

$V_D = V_T \ln\left(\dfrac{I_D}{100 I_S}\right) =$

$25 \times \ln\left(\dfrac{0.805 \times 10^{-3}}{5.3 \times 10^{-14}}\right) = 586$ mV

This is almost identical to the value computed above. Thus, the current increases by $0.805 - 0.767 = 0.038$ mA $= 38\ \mu$A.

4.39 Available diodes have 0.7 V drop at 2 mA current since $2V_D = 1.4$ V is close to 1.3 V, use N parallel pairs of diodes to split the 1 mA current evenly, as shown in the figure next.



The voltage drop across each pair of diodes is 1.3 V. ∴ Voltage drop across each diode

$= \frac{1.3}{2} = 0.65$ V. Using

$I_2 = I_1 e^{(V_2 - V_1)/V_T}$

$= 2e^{(0.65-0.7)/0.025}$

$= 0.2707$ mA

Thus current through each branch is 0.2707 mA.

The 1 mA will split in $= \frac{1}{0.2707} = 3.69$ branches.

Choose N = 4.

There are 4 pairs of diodes in parallel.

∴ We need 8 diodes.

Current through each pair of diodes

$= \frac{1 \text{ mA}}{4} = 0.25$ mA

∴ Voltage across each pair

$= 2\left[0.7 + 0.025 \ln\left(\frac{0.25}{2}\right)\right]$

$= 1.296$ V

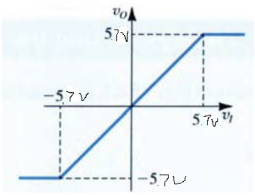
SPECIAL NOTE: There is another possible design utilizing only 6 diodes and realizing a voltage of 1.313 V. It consists of the series connection of 4 parallel diodes and 2 parallel diodes.

4.40

$v_O = -5.7$ V $v_I < -5.7$ V

$v_O = v_I$ -5.7 V $< v_I <$ +5.7 V

$v_O = 5.7$ V $v_I >$ +5.7 V



$I_{D1} = \frac{100 - 5 - 0.7}{100} = 0.94$ A

$I_{D2} = 0$

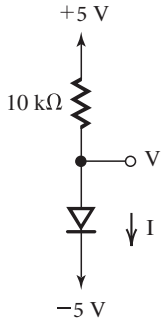
4.41

(a)

$V = -5 + 0.7 = -4.3$ V

$I = \frac{5 + 4.3}{10}$

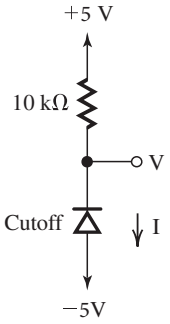
$= 0.93$ mA



(b)

$I = 0$ A

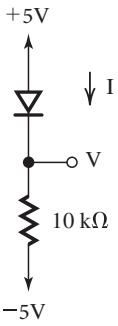
$V = 5 - I \text{ x}(10) = 5$ V



(c)

$V = 5 - 0.7 = 4.3$ V

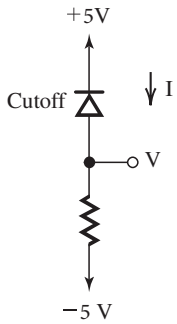
$I = \frac{4.3 + 5}{10} = 0.93$ mA



(d)

$I = 0$ A

$V = -5$ V



4.42

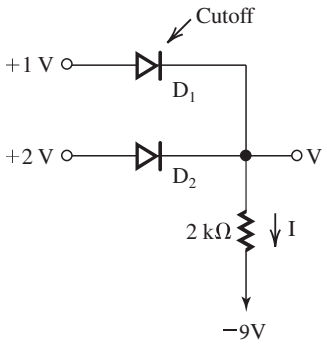
(a)

$V = 2 - 0.7$

$= 1.3$ V

$I = \dfrac{1.3 - (-9)}{2}$

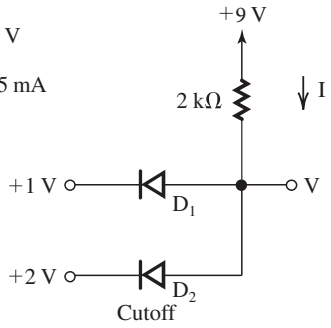
$= 5.15$ mA



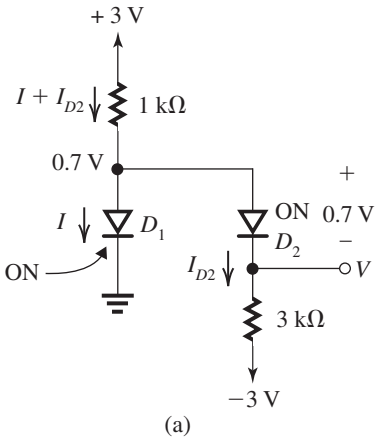
(b)

$V = 1 + 0.7 = 1.7$ V

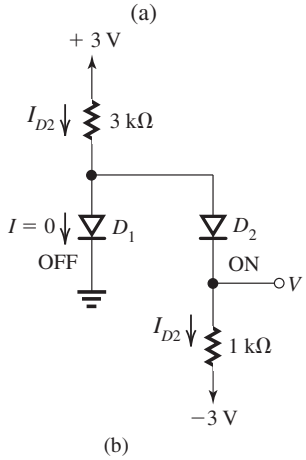
$I = \dfrac{9 - 1.7}{2} = 3.65$ mA



4.43



(a)



(b)

(a) Assuming a current flow from the positive to the negative supply voltages, D_2 will be conducting. A voltage division between the two resistors would place forward bias on D_1, so let's assume D_1 is also conducting, placing +0.7 V at its anode. Thus, as shown in Fig. (a),

$V = 0$ V

$I_{D2} = \dfrac{0 - (-3)}{3} = 1$ mA

$I + I_{D2} = \dfrac{3 - 0.7}{1} = 2.3$ mA

$\Rightarrow I = 2.3 - 1 = 1.3$ mA

These results are consistent with our assumptions.

(b) Again assuming D_2 is on, a voltage division between the two resistors will negatively bias D_1. Hence, we assume D_1 is off.

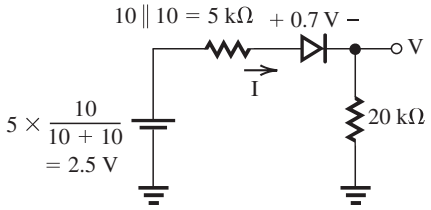
$I = 0$

$I_{D2} = \dfrac{3 - 0.7 - (-3)}{3 + 1} = 1.325$ mA

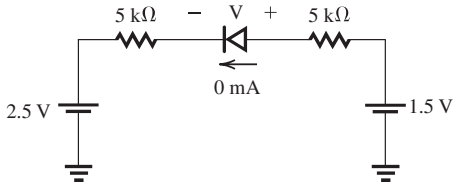
$V = -3 + 1.325 \times 1 = -1.675$ V

Thus, the anode of D_1 is at $-1.675 + 0.7 = -0.975$ V which confirms that it is reverse-biased.

4.44



(a)



(b)

(a) $I = \dfrac{2.5 - 0.7}{5 + 20} = 0.072$ mA

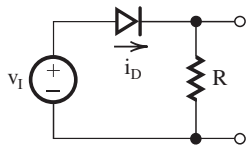
$V = 0.072 \times 20 = 1.44$ V

(b) The diode will be cut off, thus

$I = 0$

$V = 1.5 - 2.5 = -1$ V

4.45



$$i_{D,peak} = \frac{v_{I,peak} - 0.7}{R} \le 40 \text{ mA}$$

$$R \ge \frac{120\sqrt{2} - 0.7}{40} = 4.23 \text{ k}\Omega$$

Reverse voltage $= 120\sqrt{2} = 169.7$ V.

The design is essentially the same since the supply voltage $\gg 0.7$ V

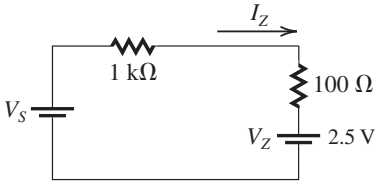
4.46 Eq. (4.8), $V_Z = V_{Z0} + r_z I_Z$, also applies to the test voltage and current. Thus,

$$V_{ZT} = V_{Z0} + r_z I_{ZT}$$

We can use this equation to complete the table as follows.

	V_{ZT}	I_{ZT}	r_z	V_{Z0}	V_Z @ $I_Z = 10$ mA
(a)	3 V	2 mA	50 Ω	**2.9 V**	**3.4 V**
(b)	5 V	5 mA	**100** Ω	4.5 V	**5.5 V**
(c)	6 V	**2 mA**	150 Ω	5.7 V	**7.2 V**
(d)	**9.1 V**	1 mA	100 Ω	9 V	**10 V**

4.47 $V_{Z0} = 3 \text{ V} - 5 \text{ m A} \times 100\Omega = 2.5$ V.



The power dissipated in the zener is

$$P_Z = V_Z I_Z = (2.5 \text{ V} + 100\Omega \times I_Z) \times I_Z$$

We use this to find the current at $P_{Z\max} = 40$ mW,

$$40 \text{ mW} = 2.5 \text{ V} \times I_{Z\max} + 100\Omega \times I_{Z\max}^2$$

$$\Rightarrow 100 I_{Z\max}^2 + 2.5 I_{Z\max} - 0.04 = 0$$

$$\Rightarrow I_{Z\max} = \frac{-2.5 + \sqrt{2.5^2 + 4 \times 100 \times 0.04}}{2 \times 100}$$

$$= 0.0111 \text{ A} = 11.1 \text{ mA}$$

This current corresponds to

$$V_S = 2.5 + 1.1 \times 11.1 = 14.71 \text{ V and}$$

$$V_Z = 2.5 + 0.1 \times 11.1 = 3.61 \text{ V}$$

4.48 The voltage change per degree Celsius of the series-connected diodes is the sum of the two diode voltage changes. Thus, the temperature coefficient is $+1.5 - 2 = -0.5$ mV/°C.

4.49 Use the exponential diode model to find the percentage change in the current.

$$i_D = I_S e^{v/V_T}$$

$$\frac{i_{D2}}{i_{D1}} = e^{(V_2 - V_1)/V_T} = e^{\Delta v/V_T}$$

For +5 mV change we obtain

$$\frac{i_{D2}}{i_{D1}} = e^{5/25} = 1.221$$

% change

$$= \frac{i_{D2} - i_{D1}}{i_{D1}} \times 100 = \frac{1.221 - 1}{1} \times 100$$

$$= 22.1\%$$

For –5 mV change we obtain

$$\frac{i_{D2}}{i_{D1}} = e^{-5/25} = 0.818$$

$$\% \text{ change} = \frac{i_{D2} - i_{D1}}{i_{D1}} \times 100 = \frac{0.818 - 1}{1} \times 100$$

$$= -18.1\%$$

Maximum allowable voltage signal change when the current change is limited to ±10% is found as follows:

The current varies from 0.9 to 1.1

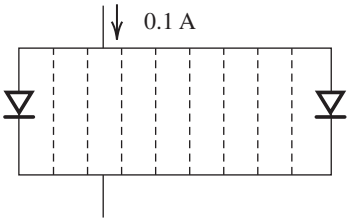
$$\frac{i_{D2}}{i_{D1}} = e^{\Delta V/V_T}$$

For 0.9, $\Delta V = 25 \ln(0.9) = -2.63$ mV

For 1.1, $\Delta V = 25 \ln(1.1) = +2.38$ mV

For ±10% current change the voltage signal change is from –2.63 mV to +2.38 mV

4.50



Ten diode connected in parallel and fed with a total current of 0.1 A. So the current through each diode $= \frac{0.1}{10} = 0.01$ A

Small signal resistance of each diode

$$= \frac{V_T}{i_D} = \frac{25 \text{ mV}}{0.01 \text{ A}} = 2.5 \ \Omega$$

Equivalent resistance, R_{eq}, of 10 diodes connected in parallel is given by

$$R_{eq} = \frac{2.5}{10} = 0.25 \ \Omega$$

If there is one diode conducting 0.1 A current, then the small signal resistance of this diode

$$= \frac{25 \text{ mV}}{0.1 \text{ A}} = 0.25 \ \Omega$$

This value is the same as of 10 diodes connected in parallel.

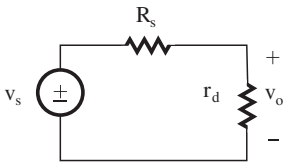
If 0.2 Ω is the resistance for making connection, the resistance in each branch
$= r_d + 0.2 = 2.5 + 0.2 = 2.7\ \Omega$

For a parallel combination of 10 diodes, equivalent resistance, R_{eq}, is

$$R_{eq} = \frac{2.7}{10} = 0.27\ \Omega$$

If there is a single diode conducting all the 0.1 A current, the connection resistance needed for the single diode will be $= 0.27 - 0.25 = 0.02\ \Omega$.

4.51 The dc current I " ows through the diode giving rise to the diode resistance $r_d = \frac{V_T}{I}$ and the small-signal equivalent circuit is represented by



$$v_o = v_s \times \frac{r_d}{r_d + R_S} = v_s \frac{V_T/I}{\frac{V_T}{I} + R_S} = v_s \frac{V_T}{V_T + IR_S}$$

Now, $v_o = 10\ \text{mV} \times \dfrac{25\ \text{mV}}{25\ \text{mV} + 10^3 I}$

I	v_o
1 mA	0.24 mV
0.1 mA	2.0 mV
1 μA	9.6 mV

For $v_o = \frac{1}{2} v_s = v_s \times \dfrac{0.025}{0.025 + 10^3 I}$

$\Rightarrow I = 25\ \mu\text{A}$

4.52 As shown in Problem 4.51,

$$\frac{v_o}{v_i} = \frac{V_T}{V_T + R_S I} = \frac{0.025}{0.025 + 10^4 I} \qquad (1)$$

Here $R_S = 10\ \text{k}\Omega$

The current changes are limited ±10%. Using exponential model, we get

$$\frac{i_{D2}}{i_{D1}} = e^{\Delta v / V_T} = 0.9 \text{ to } 1.1$$

$$\Delta v = 25 \times 10^{-3} \ln\left(\frac{i_{D2}}{i_{D1}}\right) \text{ and here}$$

For 0.9, $\Delta v = -2.63$ mV

For 1.1, $\Delta v = 2.38$ mV

The variation is –2.63 mV to 2.38 mV for ±10% current variation. Thus the largest symmetrical

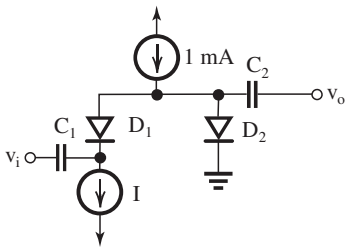
output signal allowed is 2.38 mV in amplitude. To obtain the corresponding input signal, we divide this by (v_o / v_i):

$$\hat{v}_s = \frac{2.38\ \text{mV}}{v_o / v_i} \qquad (2)$$

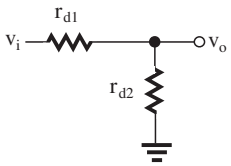
Now for the given values of (v_o / v_i) calculate I and $\hat{v}_S$ using Equations (1) and (2)

$\frac{v_o}{v_i}$	I in mA	$\hat{v}_s$ in mV
0.5	0.0025	4.76
0.1	0.0225	23.8
0.01	0.2475	238
0.001	2.4975	2380

4.53



When both D_1 and D_2 are conducting, the small-signal model is



where we have replaced the large capacitors C_1 and C_2 by short circuits:

$$\frac{v_o}{v_i} = \frac{r_{d2}}{r_{d1} + r_{d2}} = \frac{\frac{V_T}{1\ \text{m} - I}}{\frac{V_T}{I} + \frac{V_T}{1\ \text{m} - I}} = \frac{I}{1\ \text{m}}$$

Thus

$\dfrac{v_o}{v_i} = I$, where I is in mA

Now $I = 0\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 0$

$I = 1\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 1 \times 10^{-3} = 0.001\ \text{V/V}$

$I = 10\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 10 \times 10^{-3} = 0.01\ \text{V/V}$

$I = 100\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 100 \times 10^{-3} = 0.1\ \text{V/V}$

$I = 500\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 500 \times 10^{-3} = 0.5\ \text{V/V}$

$I = 600\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 600 \times 10^{-3} = 0.6\ \text{V/V}$

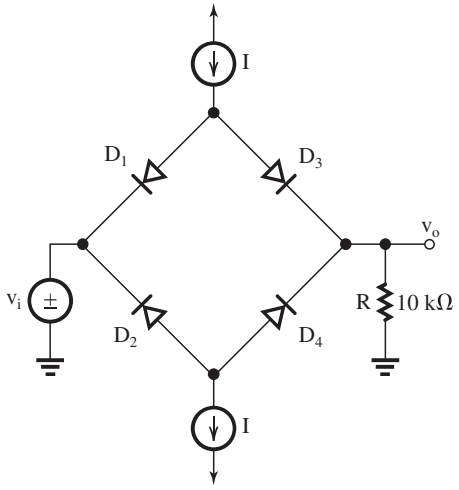
$I = 900\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 900 \times 10^{-3} = 0.9\ \text{V/V}$

$I = 990\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 990 \times 10^{-3} = 0.99\ \text{V/V}$

$I = 1\ \text{mA}$

$= 1000\ \mu\text{A}$, $\dfrac{v_o}{v_i} = 1000 \times 10^{-3} = 1\ \text{V/V}$

4.54



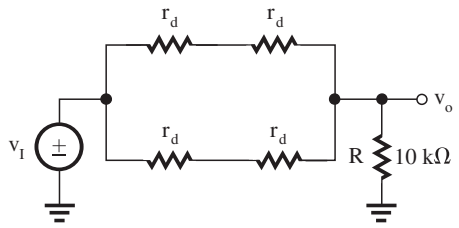
a. The current through each diode is $\frac{I}{2}$:

$$r_d = \frac{V_T}{\frac{I}{2}} = \frac{2V_T}{I} = \frac{0.05}{I}$$

From the equivalent circuit

$$\frac{v_o}{v_i} = \frac{R}{R + (2r_d \parallel 2r_d)} = \frac{R}{R + r_d}$$

I	r_d	$\frac{v_o}{v_i}$
0 μA	∞	0
1 μA	50 kΩ	0.167
10 μA	5 kΩ	0.667
100 μA	500 Ω	0.952
1 mA	50 Ω	0.995
10 mA	5 Ω	0.9995



Equivalent Circuit

b. For signal current to be limited to ±10% of I (I is the biasing current), the change in diode voltage can be obtained from the equation

$$\frac{i_D}{I} = e^{\Delta v_D / V_T} = 0.9 \text{ to } 1.1$$

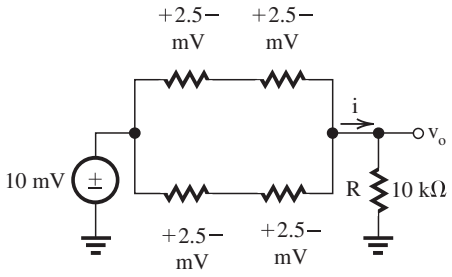
$$\Delta v_D = -2.63 \text{ mV to } +2.32 \text{ mV}$$

$$\simeq \pm 2.5 \text{ mV}$$

so the signal voltage across each diode is limited to 2.5 mV when the diode current remains within 10% of the dc bias current.

$$\therefore v_o = 10 - 2.5 - 2.5 = 5 \text{ mV}$$

$$\text{and } i = \frac{5 \text{ mV}}{10 \text{ k}\Omega} = 0.5 \text{ μA}$$



The current through each diode

$$= \frac{0.5}{2} \text{ μA} = 0.25 \text{ μA}$$

The signal current i is 0.5 μA, and this is 10% of the dc biasing current.

$\therefore$ DC biasing current $I = 0.5 \times 10 = 5$ μA

c. Now I = 1 mA.

$\therefore I_D = 0.5$ mA

Maximum current derivation 10%.

$$\therefore i_d = \frac{0.5}{10} = 0.05 \text{ mA}$$

and $i = 2i_d = 0.1$ mA.

$\therefore$ Maximum $v_o = i \times 10 \text{ k}\Omega$

$= 0.1 \times 10$

$= 1$ V

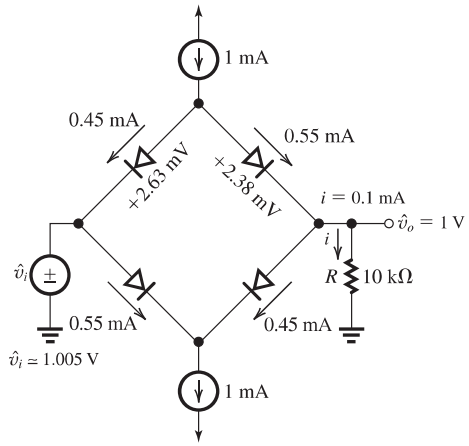
From the results of (a) above, for I = 1 mA, $v_o / v_i = 0.995$; thus the maximum input signal will be

$\hat{v}_i = \hat{v}_o / 0.995 = 1 / 0.995$

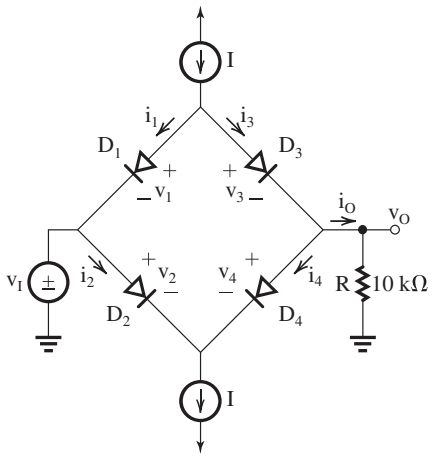
$= 1.005$ V

The same result can be obtained from the figure above where the signal across the two series diodes is 5 mV, thus

$\hat{v}_i = \hat{v}_o + 5 \text{ mV} = 1 \text{ V} + 5 \text{ mV} = 1.005$ V. See also the figure below.



4.55



$I = 1$ mA

Each diode exhibits 0.7 V drop at 1 mA current. Using diode exponential model we have

$$v_2 - v_1 = V_T \ln\left(\frac{i_2}{i_1}\right)$$

and $v_1 = 0.7$ V, $i_1 = 1$ mA

$$\Rightarrow v = 0.7 + V_T \ln\left(\frac{i}{1}\right)$$

$$= 700 + 25 \ln(i)$$

Calculation for different values of v_O:

$v_O = 0$, $i_O = 0$, and the current $I = 1$ mA divide equally between the D_3, D_4 side and the D_1, D_2 side.

$$i_1 = i_2 = i_3 = i_4 = \frac{I}{2} = 0.5 \text{ mA}$$

$$v = 700 + 25 \ln(0.5) \simeq 683 \text{ mV}$$

$$v_1 = v_2 = v_3 = v_4 = 683 \text{ mV}$$

From the circuit, we have

$$v_I = -v_1 + v_3 + v_O = -683 + 683 + 0 = 0 \text{ V}$$

For $v_O = 1$ V, $i_O = \dfrac{1}{10\,\text{k}\Omega} = 0.1$ mA

Because of symmetry of the circuit, we obtain

$$i_3 = i_2 = \frac{I}{2} + \frac{i_O}{2} = 0.5 + 0.05 = 0.55 \text{ mA and}$$

$i_4 = i_1 = 0.45$ mA

$$v_3 = v_2 = 700 + 25 \ln\left(\frac{i_2}{1}\right) = 685 \text{ mV}$$

$$v_4 = v_1 = 700 + 25 \ln\left(\frac{i_4}{1}\right) = 680 \text{ mV}$$

v_O (V)	i_O (mA)	$i_3 = i_2$ (mA)	$i_4 = i_1$ (mA)	$v_3 = v_2$ (mV)	$v_4 = v_1$ (mV)	$v_I = -v_1 + v_3 + v_O$ (V)
0	0	0.5	0.5	683	683	0
+1	0.1	0.55	0.45	685	680	1.005
+2	0.2	0.6	0.4	~687	677	2.010
+5	0.5	0.75	0.25	~693	665	5.028
+9	0.9	0.95	0.05	~699	~625	9.074
+9.9	0.99	0.995	0.005	~700	568	10.09
9.99	0.999	0.9995	0.0005	~700	510	10.18
10	1	1	0	700	0	10.7

$$v_I = -v_1 + v_2 + v_O = -0.680$$

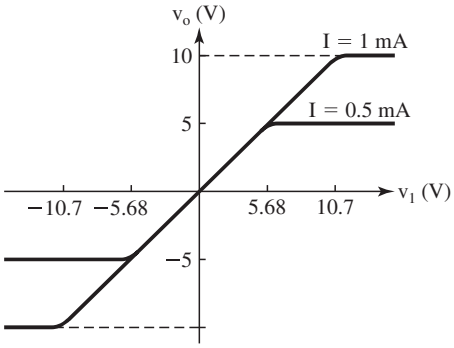
$$+0.685 + 1 = 1.005 \text{ V}$$

Similarly, other values are calculated as shown in the table. The largest values of v_O on positive and negative side are +10 V and −10 V, respectively. This restriction is imposed by the current $I = 1$ mA

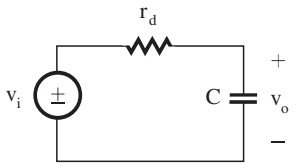
A similar table can be generated for the negative values. It is symmetrical.

For $v_I > +10.7$, v_O will be saturated at +10 V and it is because $I = 1$ mA. Similarly, for $v_I < -10.7$ V, v_O will be saturated at −10 V.

For $I = 0.5$ mA, the output will saturate at $0.5 \text{ mA} \times 10\,\text{k}\Omega = 5$ V.



4.56 Representing the diode by the small-signal resistances, the circuit is



$$r_d = \frac{V_T}{I_D}$$

$$\frac{V_o}{V_i} = \frac{\dfrac{1}{sC}}{r_d + \dfrac{1}{sC}} = \frac{1}{1 + sCr_d}$$

$$\frac{V_o}{V_i} = \frac{1}{1 + j\omega Cr_d}$$

$$\text{Phase shift} = -\tan^{-1}\left(\frac{\omega Cr_d}{1}\right)$$

$$= -\tan^{-1}\left(\omega C \frac{V_T}{I}\right)$$

For phase shift of −45°, we obtain

$$-45 = -\tan^{-1}\left(2\pi \times 100 \times 10^3 \times 10 \times 10^{-9} \times \frac{0.025}{I}\right)$$

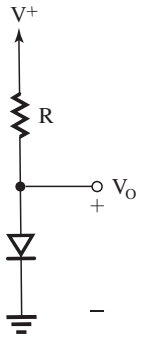
$$\Rightarrow I = 157\ \mu\text{A}$$

Now I varies from $\dfrac{157}{10}\ \mu$A to $157 \times 10\ \mu$A

Range is 15.7 μA to 1570 μA

Range of phase shift is −84.3° to −5.71°

4.57

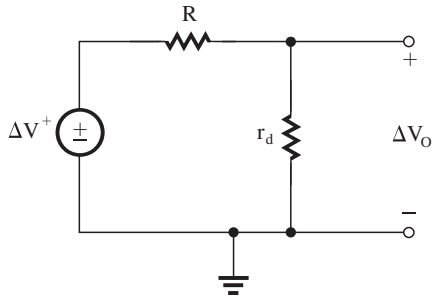


(a) $\dfrac{\Delta V_O}{\Delta V^+} = \dfrac{r_d}{R + r_d} = \dfrac{V_T / I}{R + \dfrac{V_T}{I}}$

$= \dfrac{V_T}{IR + V_T}$

For no load, $I = \dfrac{V^+ - V_O}{R} = \dfrac{V^+ - 0.7}{R}$.

$\therefore \dfrac{\Delta V_O}{\Delta V^+} = \dfrac{V_T}{V_T + (V^+ - 0.7)}$



Small-signal model

(b) If m diodes are in series, we obtain

$\dfrac{\Delta V_O}{\Delta V^+} = \dfrac{m r_d}{m r_d + R} = \dfrac{m V_T}{m V_T + IR}$

$= \dfrac{m V_T}{m V_T + (V^+ - 0.7m)}$

(c) For $m = 1$

$\dfrac{\Delta V_O}{\Delta V^+} = \dfrac{V_T}{V_T + V^+ - 0.7}$

$= \dfrac{25}{0.025 + 10 - 0.7}$

$= 2.68$ mV/ V

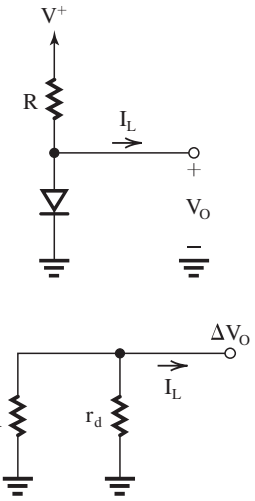
For $m = 3$

$\dfrac{\Delta V_O}{\Delta V^+} = \dfrac{m V_T}{m V_T + 10 - m \times 0.7}$

$= \dfrac{3 \times 25}{3 \times 0.025 + 10 - 3 \times 0.7}$

$= 9.40$ mV/ V

4.58



Small-signal model

(a) From the small-signal model

$\Delta V_O = -I_L (r_d \parallel R)$

$\dfrac{\Delta V_O}{I_L} = -(r_d \parallel R)$

(b) At no load, $I_D = \dfrac{V^+ - 0.7}{R}$

$r_d = \dfrac{V_T}{I_D}$

$\dfrac{\Delta V_O}{I_L} = -(r_d \parallel R) = -\dfrac{1}{\dfrac{1}{r_d} + \dfrac{1}{R}}$

$= -\dfrac{1}{\dfrac{I_D}{V^+ - 0.7} + \dfrac{I_D}{V_T}}$

$= -\dfrac{V_T}{I_D} \times \dfrac{1}{\dfrac{V_T}{V^+ - 0.7} + 1}$

$= -\dfrac{V_T}{I_D} \times \dfrac{V^+ - 0.7}{V_T + V^+ - 0.7}$

For $\dfrac{\Delta V_O}{I_L} \le 5 \dfrac{\text{mV}}{\text{mA}}$

i.e., $\dfrac{V_T}{I_D} \times \dfrac{V^+ - 0.7}{V_T + V^+ - 0.7} \le \dfrac{5 \text{ mV}}{\text{mA}}$

$\left[\dfrac{25}{I_D} \times \dfrac{15 - 0.7}{0.025 + 15 - 0.7}\right] \le 5 \dfrac{\text{mV}}{\text{mA}}$

$I_D \ge 4.99$ mA

$I_D \simeq 5$ mA

$R = \dfrac{V^+ - 0.7}{I_D} = \dfrac{15 - 0.7}{5 \text{ mA}}$

$R = 2.86$ kΩ

Diode should be a 5-mA unit; that is, it conducts 5 mA at $V_D = 0.7$ V, thus $5 \times 10^{-3} = I_S e^{0.7/0.025}$.

$\Rightarrow I_S = 3.46 \times 10^{-15}$ A

(c) For m diodes connected in series we have

$$I_D = \frac{V^+ - 0.7m}{R}$$

and $r_d = \frac{V_T}{I_D}$

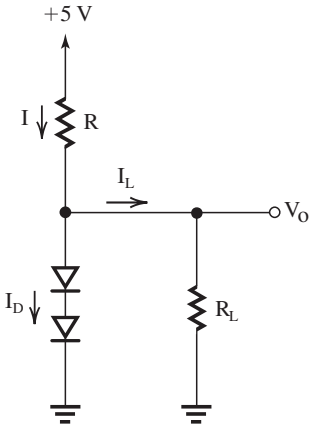
So now $$\frac{\Delta V_O}{I_L} = -(R \parallel mr_d) = -\frac{1}{\frac{1}{R} + \frac{1}{mr_d}}$$

$$= -\frac{1}{\frac{I_D}{V^+ - 0.7m} + \frac{I_D}{mV_T}}$$

$$= -\frac{mV_T}{I_D}\frac{}{\frac{mV_T}{V^+ - 0.7m} + 1}$$

$$= -\frac{mV_T}{I_D}\frac{V^+ - 0.7m}{V^+ - 0.7m + mV_T}$$

4.59



Diode has 0.7 V drop at 1 mA current

$V_O = 1.5$ V when $R_L = 1.5$ kΩ

$I_D = I_S e^{V/V_T}$

$1 \times 10^{-3} = I_S e^{0.7/0.025}$

$\Rightarrow I_S = 6.91 \times 10^{-16}$ A

Voltage drop across each diode = $\frac{1.5}{2} = 0.75$ V.

$\therefore I_D = I_S e^{V/V_T} = 6.91 \times 10^{-16} \times e^{0.75/0.025}$

$= 7.38$ mA

$I_L = 1.5/1.5 = 1$ mA

$I = I_D + I_L = 7.39$ mA $+ 1$ mA

$= 8.39$ mA

$$\therefore R = \frac{5 - 1.5}{8.39 \text{ mA}} = 417\ \Omega$$

$P_D = I_D$ x 1.5V=11.09 mW

$P_L = I_L$ x 1.5V=1.5mW

Use a small-signal model to find voltage ΔV_O when the value of the load resistor, R_L, changes:

$$r_d = \frac{V_T}{I_D} = \frac{0.025}{7.39} = 3.4\ \Omega$$

When load is disconnected, all the current I flows through the diode. Thus

$\Delta I_D = 1$ mA

$\Delta V_O = \Delta I_D \times 2r_d$

$= 1 \times 2 \times 3.4$

$= 6.8$ mV

With $R_L = 1$ kΩ,

$$I_L \simeq \frac{1.5 \text{ V}}{1} = 1.5 \text{ mA}$$

$\Delta I_L = 0.5$ mA

$\Delta I_D = -0.5$ mA

$\Delta V_O = -0.5 \times 2 \times 3.4$

$= -3.4$ mV

With $R_L = 750\ \Omega$,

$$I_L \simeq \frac{1.5}{0.75} = 2 \text{ mA}$$

$\Delta I_L = 1$ mA

$\Delta I_D = -1$ mA

$\Delta V_O = -1 \times 2 \times 3.4$

$= -6.8$ mV

With $R_L = 500\ \Omega$,

$$I_L \simeq \frac{1.5}{0.5} = 3 \text{ mA}$$

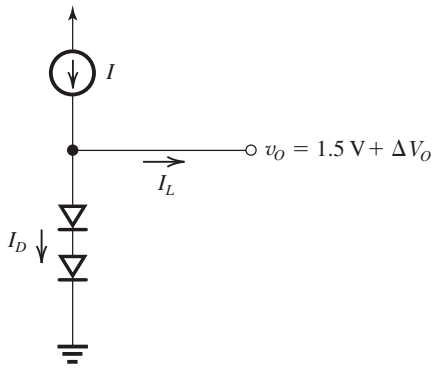
$\Delta I_L = 2.0$ mA

$\Delta I_D = -2.0$ mA

$\Delta V_O = -2 \times 2 \times 3.4$

$= -13.6$ mV

4.60



I_L varies from 2 to 7 mA

To supply a load current of 2 to 7 mA, the current I must be greater than 7 mA. So I can be only 10 mA or 15 mA.

Now let us evaluate ΔV_O for both 10-mA and 15-mA sources. For the 10-mA source:

Since I_L varies from 2 to 7 mA, the current I_D will varry from 8 to 3 mA.

Correspondingly, the voltage across each diode changes by $\triangle V_D$ where

$$\frac{3}{8} = e^{\triangle V_D/V_T}$$

$$\Rightarrow \triangle V_D = 25 \ln\left(\frac{3}{8}\right) = -24.5 \text{ mV}$$

and the output voltage changes by

$$\triangle V_O = 2 \times \triangle V_D = -49 \text{ mV}$$

With $I = 15$ mA, the diodes current changes from 13 to 8 mA. Correspondingly, the voltage across each diode changes by $\triangle V_D$ where

$$\frac{8}{13} = e^{\triangle V_D/V_T}$$

$$\Rightarrow \triangle V_D = 25 \ln\left(\frac{8}{13}\right) = -12.1 \text{ mV}$$

and the output voltage changes by

$$\triangle V_O = 2 \times \triangle V_D = -24.2 \text{ mV}$$

which is less than half that obtained with the 10-mA supply. Thus, from the point of view of reducing the change in V_O as I_L changes, we choose the 15-mA supply. Note, however, that the price paid is increased power dissipation.

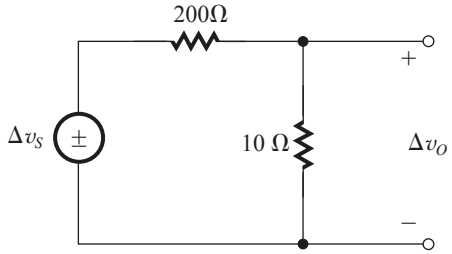
4.61 (a) Three 6.8-V zeners provide $3 \times 6.8 = 20.4$ V with $3 \times 10 = 30$-Ω resistance. Neglecting R, we have

Load regulation $= -30$ mV/mA.

(b) For 5.1-V zeners we use 4 diodes to provide 20.4 V with $4 \times 30 = 120$-Ω resistance.

Load regulation $= -120$ mV/mA

4.62



Small-signal model

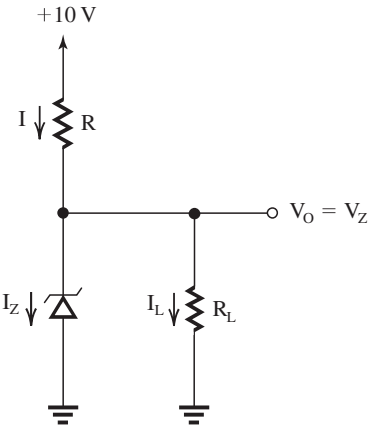
From the small-signal model we obtain

$$\frac{\Delta v_O}{\Delta v_S} = \frac{10}{10 + 200} = \frac{1}{21}$$

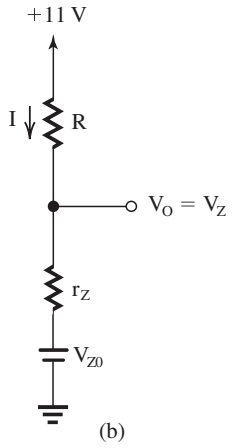
Now $\Delta v_S = 1.0$ V.

$$\therefore \Delta v_O = \frac{1}{21}\Delta v_S = \frac{1}{21} \times 1.0 = 47.6 \text{ mV}$$

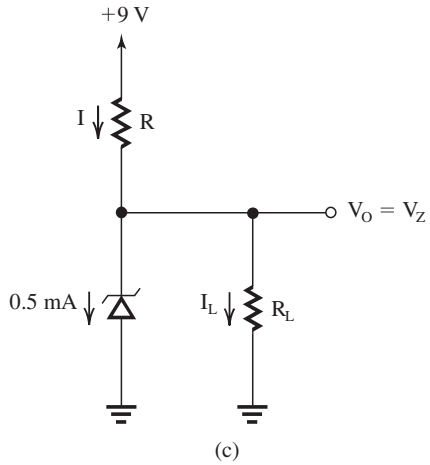
4.63



(a)



(b)



(c)

To obtain $V_O = 7.5$ V, we must arrange for $I_Z = 10$ mA (the current at which the zener is specified).

Now,

$$R_L = \frac{V_O}{I_L} = \frac{7.5}{5} = 1.5\text{k}\Omega$$

Thus

$$I = I_Z + I_L = 10 + 5 = 15 \text{ mA}$$

and

$$R = \frac{10 - V_O}{I} = \frac{10 - 7.5}{15} = 167\ \Omega$$

When the supply undergoes a change $\triangle V_S$, the change in the output voltage, $\triangle V_O$, can be determined from

$$\frac{\triangle V_O}{\triangle V_S} = \frac{(R_L \parallel r_z)}{(R_L \parallel r_z) + R}$$

$$= \frac{1.5 \parallel 0.03}{(1.5 \parallel 0.03) + 0.167} = 0.15$$

For $\triangle V_S = +1$ V (10% high), $\triangle V_O = +0.15$ V and $V_O = 7.65$ V.

For $\triangle V_S = -1$ V (10% low), $\triangle V_O = -0.15$ V and $V_O = 7.35$ V.

When the load is removed and $V_S = 11$ V, we can use the zener model to determine V_O. Refer to Fig. (b). To determine V_{Z0}, we use

$$V_Z = V_{Z0} + I_{ZT} r_z$$

$$7.5 = V_{Z0} + 0.01 \times 30$$

$$\Rightarrow V_{Z0} = 7.2 \text{ V}$$

From Fig. (b) we have

$$I = \frac{11 - 7.2}{0.167 + 0.03} = 19.3 \text{ mA}$$

Thus

$$V_O = V_{Z0} + I r_z$$

$$= 7.2 + 0.0193 \times 30 = 7.78 \text{ V}$$

To determine the smallest allowable value of R_L while $V_S = 9$ V, refer to Fig. (c). Note that $I_Z = 0.5$ mA, thus

$$V_Z = V_{ZK} \simeq V_{Z0} = 7.2 \text{ V}$$

$$I = \frac{9 - 7.2}{0.167} = 10.69 \text{ mA}$$

$$I_L = I - I_Z = 10.69 - 0.5 = 10.19 \text{ mA}$$

$$R_L = \frac{V_O}{I_L} = \frac{7.2}{10.19} = 707\ \Omega$$

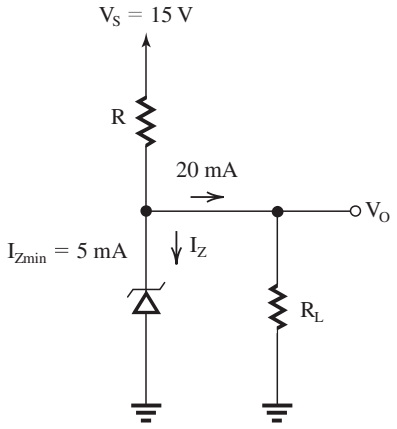
I_l=10.19mA V_O=7.2 V

4.64 (a) $V_{ZT} = V_{Z0} + r_z I_{ZT}$

$10 = V_{Z0} + 7\,(0.025)$

$\Rightarrow V_{Z0} = 9.825$ V

(b) The minimum zener current of 5 mA occurs when $I_L = 20$ mA and V_S is at its minimum of $20(1 - 0.25) = 15$ V. See the circuit below:



$$R \le \frac{15 - V_{Z0}}{20 + 5}$$

where we have used the minimum value of V_S, the maximum value of load current, and the minimum required value of zener diode current, and we assumed that at this current $V_Z \simeq V_{Z0}$. Thus,

$$R \le \frac{15 - 9.825 + 7}{25}$$

$\le 207\ \Omega$.

$\therefore$ use $R = 207\ \Omega$

(c) Line regulation $= \dfrac{7}{207 + 7} = 33\ \dfrac{\text{mV}}{\text{V}}$

$\pm 25\%$ change in $v_S \equiv \pm 5$ V

V_O changes by $\pm 5 \times 33 = \pm 0.165$ mV

corresponding to $\dfrac{\pm 0.165}{10} \times 100 = \pm 1.65\%$

(d) Load regulation $= -(r_Z \parallel R)$

$= -(7 \parallel 207) = -6.77\ \Omega$

or –6.77 V/A

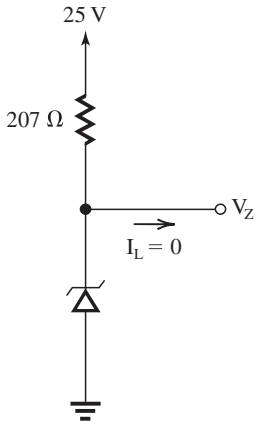
$\Delta V_O = -6.77 \times 20\text{ mA} = -135.4$ mV

corresponding to $-\dfrac{0.1354}{10} \times 100 = -1.35\%$

(e) The maximum zener current occurs at no load $I_L = 0$ and the supply at its largest value of

$20 + \dfrac{1}{4}(20) = 25$ V.

$V_Z = V_{Z0} + r_Z I_Z$



$$= 9.825 + 7 \times \frac{25 - V_Z}{207}$$

$207 V_Z = 207\,(9.825) + 7\,(25) - 7 V_Z$

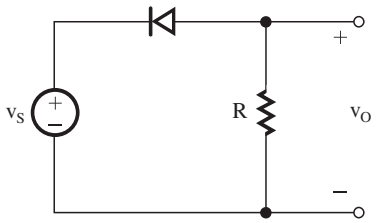
$\Rightarrow V_Z = 10.32$ V

$I_{Zmax} = \dfrac{25 - 10.32}{0.207} = 70.9$ mA

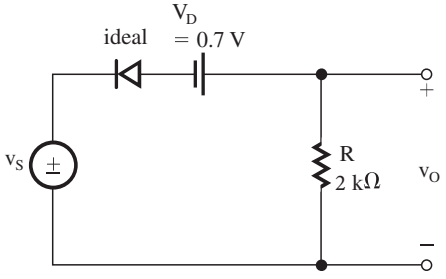
$P_Z = 10.32 \times 70.9$

$= 732$ mW

4.65

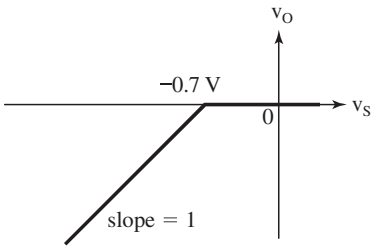


Using the constant voltage drop model:

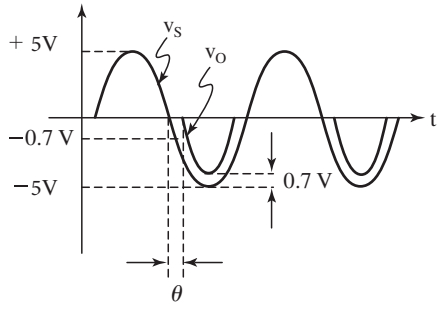


(a) $v_O = v_S + 0.7$ V, For $v_S \leq -0.7$ V

$v_O = 0$, for $v_S \geq -0.7$ V



(b)



(c) The diode conducts at an angle

$\theta = \sin^{-1}\left(\frac{0.7}{5}\right) = 8^\circ$ and stops

at $\pi - \theta = 172^\circ$

Thus the conduction angle is $\pi - 2\theta$

$= 164^\circ$ or 2.86 rad.

$$v_{O,avg} = \frac{-1}{2\pi}\int_{\theta}^{\pi-\theta} (5\sin\phi - 0.7)\,d\phi$$

$$= \frac{-1}{2\pi}[-5\cos\phi - 0.7\phi]_{\theta}^{\pi-\theta}$$

$= -1.26$ V

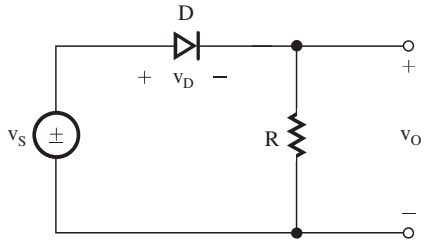
(d) Peak current in diode is

$$\frac{5 - 0.7}{2} = 2.15 \text{ mA}$$

(e) PIV occurs when v_S is at its the peak and $v_O = 0$.

PIV= 5 V

4.66



$i_D = I_S e^{v_D/V_T}$

$$\frac{i_D}{i_D\,(1\text{ mA})} = e^{[V_D - V_{D(\text{at 1 mA})}]/V_T}$$

$$v_D - v_D(\text{at 1 mA}) = V_T \ln\left[\frac{i_D}{1\text{ mA}}\right]$$

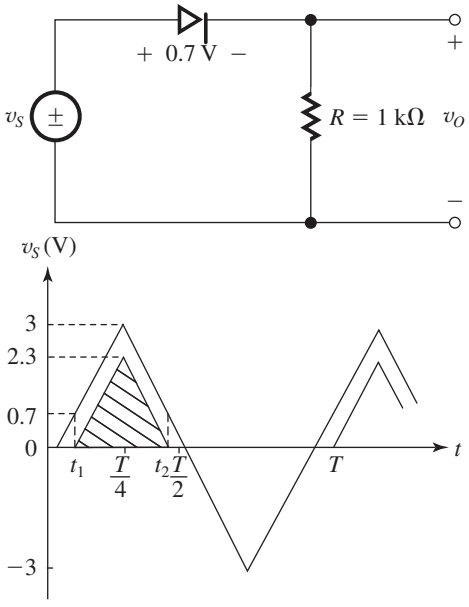
$$v_D = v_D(\text{at 1 mA}) + V_T \ln\left[\frac{v_O/R}{1}\right]$$

$v_O = v_S - v_D$

$$= v_S - v_D\,(\text{at 1 mA}) - V_T \ln\left(\frac{v_O}{R}\right)$$

where R is in kΩ.

4.67



First find t_1 and t_2

$$\frac{3}{\frac{T}{4}} = \frac{0.7}{t_1}$$

$\Rightarrow t_1 = 0.0583$ T

$$t_2 = \frac{T}{2} - t_1$$

$$= \frac{T}{2} - 0.0583\ \text{T}$$

$$t_2 = 0.442\,\text{T}$$

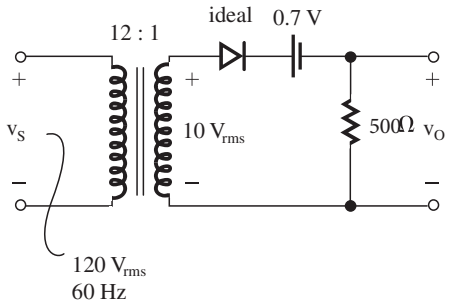
$$v_O(\text{ave.}) = \frac{1}{T} \times \text{ area of shaded triangle}$$

$$= \frac{1}{T} \times (3 - 0.7) \times \left(\frac{T}{4} - t_1\right)$$

$$= \frac{1}{T} \times 2.3 \times T\left(\frac{1}{4} - 0.0583\right)$$

$$= 0.441\ \text{V}$$

4.68



$$\hat{v}_O = 10\sqrt{2} - 0.7 = 13.44\ \text{V}$$

Conduction begins at

$$10\sqrt{2}\sin\theta = 0.7$$

$$\theta = \sin^{-1}\left(\frac{0.7}{10\sqrt{2}}\right) = 2.84°$$

$$= 0.0495\ \text{rad}$$

Conduction ends at $\pi - \theta$.

$\therefore$ Conduction angle $= \pi - 2\theta = 3.04$ rad

The diode conducts for

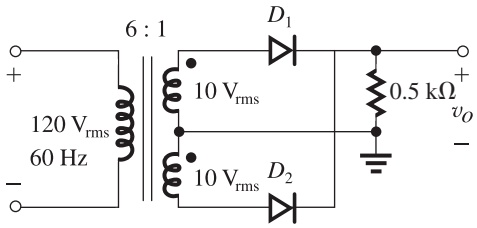
$$\frac{3.04}{2\pi} \times 100 = 48.4\% \text{ of the cycle}$$

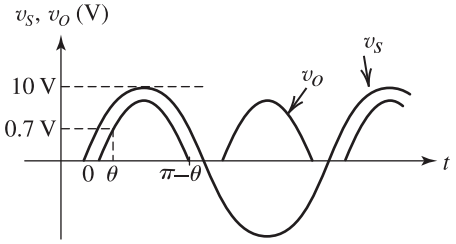
$$v_{O,\text{avg}} = \frac{1}{2\pi}\int_{\theta}^{\pi-\theta} (10\sqrt{2}\sin\phi - 0.7)\,d\phi$$

$$= 4.15\ \text{V}$$

$$i_{D,\text{avg}} = \frac{v_{O,\text{avg}}}{R} = 8.3\ \text{mA}$$

4.69





$$\hat{v}_O = 10\sqrt{2} - V_D = 13.44\ \text{V}$$

Conduction starts at $\theta = \sin^{-1}\dfrac{0.7}{10\sqrt{2}} =$

$2.84° = 0.05$ rad

and ends at $\pi - \theta$. Conduction angle $=$ $\pi - 2\theta = 3.04$ rad in each half cycle. Thus the

fraction of a cycle for which one of the two diodes conduct $= \dfrac{2(3.04)}{2\pi} \times 100 = 96.8\%$

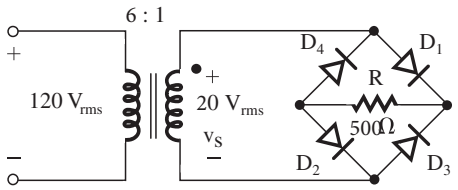
Note that during 96.8% of the cycle there will be conduction. However, each of the two diodes conducts for only half the time, i.e., for 48.4% of the cycle.

$$v_{O,\text{avg}} = \frac{1}{\pi}\int_{\theta}^{\pi-\theta} (10\sqrt{2}\sin\phi - 0.7)d\phi$$

$$= 8.3\ \text{V}$$

$$i_{L,\text{avg}} = \frac{8.3}{0.5\ \text{k}\Omega} = 16.6\ \text{mA}$$

4.70

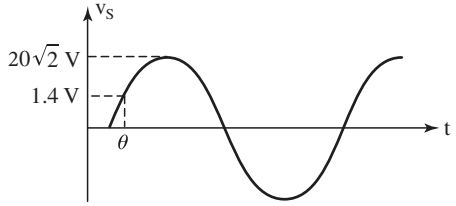


$V_D = 0.7$ V

Peak voltage across R $= 20\sqrt{2} - 2V_D$

$= 20\sqrt{2} - 1.4$

$= 26.88$V



$$\theta = \sin^{-1}\frac{1.4}{20\sqrt{2}} = 2.84^\circ = 0.05 \text{ rad}$$

Fraction of cycle that D_1 & D_2 conduct is

$$\frac{\pi - 2\theta}{2\pi} \times 100 = 48.4\%$$

Note that D_3 & D_4 conduct in the other half cycle so that there is 2 (48.4) $= 96.8\%$ conduction interval.

$$v_{O,avg} = \frac{2}{2\pi}\int_{\theta}^{\pi-\theta}(20\sqrt{2}\sin\phi - 2V_D)\,d\phi$$

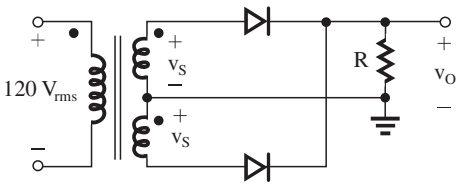
$$= \frac{1}{\pi}\left[-20\sqrt{2}\cos\phi - 1.4\phi\right]_{\theta}^{\pi-\theta}$$

$$= \frac{2(20\sqrt{2}\cos\theta)}{\pi} - \frac{1.4(\pi - 2\theta)}{\pi}$$

$= 16.63$ V

$$i_{R,avg} = \frac{v_{O,avg}}{R} = \frac{16.63}{0.5} = 33.26 \text{ mA}$$

4.71



Refer to Fig. 4.22(a).

For $V_D \ll V_s$, conduction angle $\simeq \pi$, and

$$v_{O,avg} = \frac{2}{\pi}V_s - V_D = \frac{2}{\pi}V_s - 0.7$$

(a) For $v_{O,avg} = 10$ V

$$V_s = \frac{\pi}{2} \times 10.7 = 16.8 \text{ V}$$

$$\text{Turns ratio} = \frac{120\sqrt{2}}{16.8} = 10.1 \text{ to } 1$$

(b) For $v_{O,avg} = 25$ V

$$V_s = \frac{\pi}{2} \times 25.7 = 40.37 \text{ V}$$

$$\text{Turns ratio} = \frac{120\sqrt{2}}{40.37} = 4.2 \text{to } 1$$

(c) For $v_{op} = 20$ V

$V_s = 20+0.7$ V

$$\text{Turns ratio} = \frac{120\sqrt{2}}{20.7} = 8.2 \text{to } 1$$

4.72 Refer to Fig. 4.23(a)

For $2V_D \ll V_s$

$$V_{O,avg} = \frac{2}{\pi}V_s - 2V_D = \frac{2}{\pi}V_s - 1.4$$

(a) For $V_{O,avg} = 10$ V

$$10 \text{ V} = \frac{2}{\pi} \cdot V_s - 1.4$$

$$\therefore \hat{V}_s = 11.4\left(\frac{\pi}{2}\right) = 17.9 \text{ V}$$

$$\text{Turns ratio} = \frac{120\sqrt{2}}{17.9} = 9.5 \text{ to } 1$$

(b) For $V_{O,avg} = 25$V

$$25 \text{ V} = \frac{2}{\pi} \cdot V_s - 1.4$$

$$\Rightarrow V_s = 26.4\left(\frac{\pi}{2}\right) = 41.5\text{V}$$

$$\text{Turns ratio} = \frac{120\sqrt{2}}{41.5} = 4.1 \text{ to } 1$$

(c) For $v_{op} = 20$ V

$V_s = 20+ 2x0.7$ V

$$\text{Turns ratio} = \frac{120\sqrt{2}}{21.4} = 7.9 \text{to } 1$$

4.73 $120\sqrt{2} \pm 10\%$: $20\sqrt{2} \pm 10\%$

$\Rightarrow$ Turns ratio = 6:1

$$v_S = \frac{20\sqrt{2}}{2} \pm 10\%$$

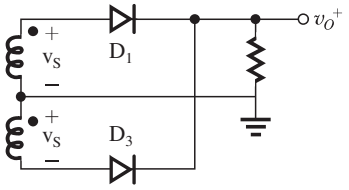
$$\text{PIV} = 2V_s - V_D$$

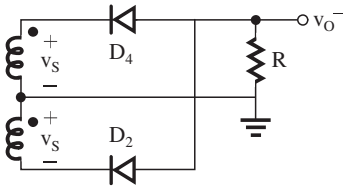
$$= 2 \times \frac{20\sqrt{2}}{2} \times 1.1 - 0.7$$

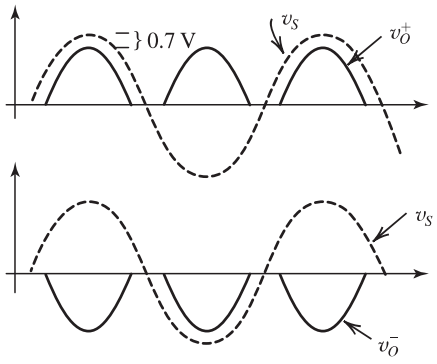
$$= 30.4 \text{ V}$$

Using a factor of 1.5 for safety, we select a diode having a PIV rating of approximately 45 V.

4.74 The circuit is a full-wave rectifier with center tapped secondary winding. The circuit can be analyzed by looking at v_O^+ and v_O^- separately.







$$v_{O,\,avg} = \frac{1}{2\pi}\int (V_S \sin\phi - 0.7)\, d\phi = 12$$

$$= \frac{2V_s}{\pi} - 0.7 = 12$$

where we have assumed $V_s \gg 0.7$ V and thus the conduction angle (in each half cycle) is almost π.

$$V_s = \frac{12 + 0.7}{2}\pi = 19.95 \text{ V}$$

Thus voltage across secondary winding

$$= 2V_S \simeq 40 \text{ V}$$

Looking at D_4,

$$\text{PIV} = V_S - V_O^-$$

$$= V_S + (V_S - 0.7)$$

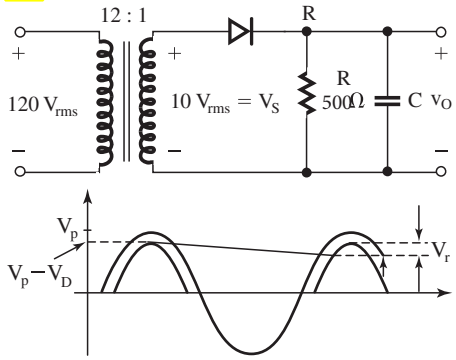
$$= 2V_S - 0.7$$

$$= 39.2 \text{ V}$$

If choosing a diode, allow a safety margin by moving a factor of 1.5, thus

$$\text{PIV} \simeq 60 \text{ V}$$

4.75



(i) $V_r \cong (V_p - V_D)\dfrac{T}{CR}$ [Eq. (4.28)]

$0.1(V_p - V_D) = (V_p - V_D)\dfrac{T}{CR}$

$C = \dfrac{1}{0.1 \times 60 \times 500} = 333.3\,\mu F$

(ii) For

$V_r = 0.01(V_p - V_D) = \dfrac{(V_p - V_D)T}{CR}$

$C = 3333\mu F$

(a) (i) $v_{O,\,avg} = V_p - V_D - \dfrac{1}{2}V_r$

$= 10\sqrt{2} - 0.7 - \dfrac{1}{2}\left(10\sqrt{2} - 0.7\right)0.1$

$= \left(10\sqrt{2} - 0.7\right)\left(1 - \dfrac{0.1}{2}\right)$

$= 12.77$ V

(ii) $v_{O,\,avg} = \left(10\sqrt{2} - 0.7\right)\left(1 - \dfrac{0.01}{2}\right)$

$= 13.37$ V

(b) (i) Using Eq (3.30), we have the conduction angle =

$\omega\Delta t \cong \sqrt{2V_r/(V_p - V_D)}$

$= \sqrt{\dfrac{2 \times 0.1(V_p - 0.7)}{(V_p - 0.7)}}$

$= \sqrt{0.2}$

$= 0.447$ rad

$\therefore$ Fraction of cycle for

conduction $= \dfrac{0.447}{2\pi} \times 100$

$= 7.1\%$

(ii) $\omega\Delta t \simeq \sqrt{2 \times 0.01\dfrac{(V_p - 0.7)}{V_p - 0.7}} = 0.141$ rad

Fraction of cycle $= \dfrac{0.141}{2\pi} \times 100 = 2.24\%$

(c) (i) Use Eq (4.31):

$i_{D,avg} = I_L\left(1 + \pi\sqrt{\dfrac{2(V_p - V_D)}{V_r}}\right)$

$= \dfrac{v_{O,avg}}{R}\left(1 + \pi\sqrt{\dfrac{2(V_p - V_D)}{0.1(V_p - V_D)}}\right)$

$= \dfrac{12.77}{0.5}\left(1 + \pi\sqrt{\dfrac{2}{0.1}}\right)$

$= 384$mA

(ii) $i_{D,avg} = \dfrac{13.37}{0.5}\left(1 + \pi\sqrt{200}\right)$

$= 1214$mA

(d) Adapting Eq. (3.32), we obtain

(i) $i_{D,peak} = I_L\left(1 + 2\pi\sqrt{\dfrac{2(V_p - V_D)}{V_r}}\right)$

$= \dfrac{12.77}{0.5}\left(1 + 2\pi\sqrt{\dfrac{2}{0.1}}\right)$

$= 742$ mA

(ii) $i_{D,peak} = \dfrac{13.37}{0.5}\left(1 + 2\pi\sqrt{\dfrac{2}{0.01}}\right)$

$= 2402$ mA $\simeq 2.4$ A

4.76 (i) $V_r = 0.1\,(V_p - V_D) = \dfrac{(V_p - V_D)}{2fCR}$

The factor of 2 accounts for discharge occurring only during half of the period, $T/2 = \dfrac{1}{2f}$.

$$C = \frac{1}{(2fR)\,0.1} = \frac{1}{2\,(60)\,x\,500\,x0.1} = 166.6\,\mu F$$

(ii) $$C = \frac{1}{2\,(60)\,x\,500\,x0.01} = 1666\,\mu F$$

(a) (i) $$V_O = V_p - V_D - \frac{1}{2}V_r$$

$$= (V_p - V_D)\left(1 - \frac{0.1}{2}\right)$$

$$= (13.44)\left(1 - \frac{0.1}{2}\right)$$

$$= 12.77\ V$$

(ii) $$V_O = (13.44)\left(1 - \frac{0.01}{2}\right) = 13.37\ V$$

(b) (i) Fraction of cycle $= \dfrac{2\omega\Delta t}{2\pi} \times 100$

$$= \frac{\sqrt{2V_r/\,(V_p - V_D)}}{\pi} \times 100$$

$$= \frac{1}{\pi}\sqrt{2\,(0.1)} \times 100 = 14.2\%$$

(ii) Fraction of cycle $= \dfrac{2\sqrt{2\,(0.01)}}{2\pi} \times 100$

$= 4.5\%$

(c) Use Eq. (4.34):

(i) $$i_{D,\,avg} = I_L\left(1 + \pi\sqrt{\frac{V_p - V_D}{2V_r}}\right)$$

$$= \frac{12.77}{0.5}\left(1 + \pi\sqrt{\frac{1}{2\,(0.1)}}\right) = 205\ mA$$

(ii) $$i_{D,\,avg} = \frac{13.37}{0.5}\left(1 + \pi\frac{1}{\sqrt{2\,(0.01)}}\right)$$

$= 620\ mA$

(d) Use Eq. (3.35):

(i) $$\hat{i}_D = I_L\left(1 + 2\pi\frac{1}{\sqrt{2\,(0.1)}}\right) = 384\ mA$$

(ii) $$\hat{i}_D = I_L\left(1 + 2\pi\frac{1}{\sqrt{0.02}}\right) = 1214\ mA$$

4.77 (i) $V_r = 0.1\,(V_p - V_D \times 2) = \dfrac{V_p - 2V_D}{2fCR}$

$$C = \frac{(V_p - 2V_D)}{(V_p - 2V_D)}\frac{1}{2\,(0.1)\,fR} = 166.6\mu F$$

(ii) $$C = \frac{1}{2\,(0.01)\,fR} = 1666\,\mu F$$

(a) $$V_O = V_p - 2V_D - \frac{1}{2}V_r$$

(i) $$V_O = V_p - 2V_D - \frac{1}{2}\,(V_p - 2V_D) \times 0.1$$

$$= (V_p - 2V_D) \times 0.95$$

$$= (10\sqrt{2} - 2 \times 0.7) \times 0.95 = 12.1\ V$$

(ii) $V_O = (10\sqrt{2} - 2 \times 0.7) \times 0.995 = 12.68\ V$

(b) (i) Fraction of cycle $= \dfrac{2\omega\Delta t}{2\pi} \times 100$

$$= \frac{\sqrt{2\,(0.1)}}{\pi} \times 100 = 14.2\%$$

(ii) Fraction of cycle $= \dfrac{\sqrt{2\,(0.01)}}{\pi} \times 100 = 4.5\%$

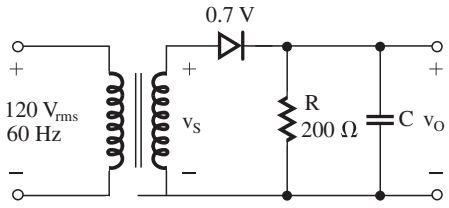
(c) (i) $$i_{D,\,avg} = \frac{12.1}{0.5}\left(1 + \pi\sqrt{\frac{1}{0.2}}\right) = 194\ mA$$

(ii) $$i_{D,\,avg} = \frac{12.68}{0.5}\left(1 + \pi/\sqrt{0.02}\right) = 498\ mA$$

(d) (i) $$\hat{i}_D = \frac{12.1}{0.5}\left(1 + 2\pi\sqrt{\frac{1}{0.2}}\right) = 364\ mA$$

(ii) $$\hat{i}_D = \frac{12.68}{0.5}\left(1 + 2\pi\sqrt{\frac{1}{0.02}}\right) = 1152mA$$

4.78



$V_O = 12\ V \pm 1\ V$ (ripple)

$R_L = 200\ \Omega$

(a) $V_O = V_p - V_D - 1$

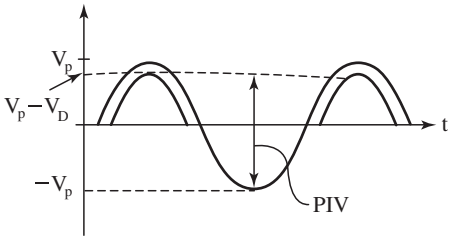
$\Rightarrow V_p = 13 + 0.7 = 13.7\ V$

$$V_{rms} = \frac{13.7}{\sqrt{2}} = 9.7\ V$$

(b) $$V_r = \frac{V_p - V_D}{fCR}$$

$$2 = \frac{13.7 - 0.7}{60 \times C \times 200}$$

$$\Rightarrow C = \frac{13}{2 \times 60 \times 200} = 542\ \mu F$$



(c) When the diode is cut off, the maximum reverse voltage across it will occur when $v_S = -V_p$. At this time, $v_O = V_O$ and the maximum reverse voltage will be

Maximum reverse voltage $= V_O + V_p$

$= 12 + 13.7 = 25.7$ V

Using a factor of safety of 1.5 we obtain

PIV $= 1.5 \times 25.7$

$= 38.5$ V

(d) $i_{Dav} = I_L\left[1 + \pi\sqrt{\dfrac{2(V_p - V_D)}{V_r}}\right]$

$= \dfrac{V_O}{R_L}\left[1 + \pi\sqrt{\dfrac{2(V_p - V_D)}{V_r}}\right]$

$= \dfrac{12}{0.2}\left[1 + \pi\sqrt{\dfrac{2(13.7 - 0.7)}{2}}\right]$

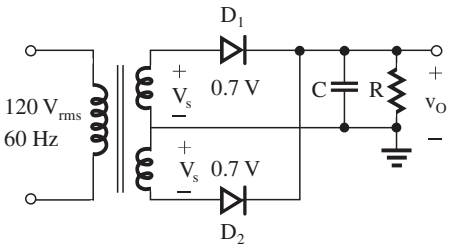
$= 739$ mA

(e) $i_{Dmax} = I_L\left[1 + 2\pi\sqrt{\dfrac{2(V_p - V_D)}{V_r}}\right]$

$= \dfrac{12}{0.2}\left[1 + 2\pi\sqrt{\dfrac{2(13.7 - 0.7)}{2}}\right]$

$= 1.42$ A

4.79

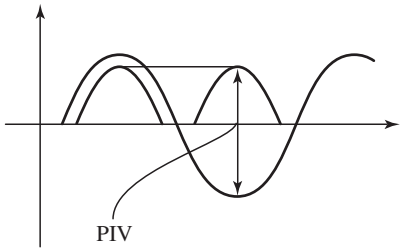


(a) $V_O = V_p - V_D - 1$

$\Rightarrow V_p = V_O + V_D + 1 = 13 + 0.7 = 13.7$ V

$V_{rms} = \dfrac{13.7}{\sqrt{2}} = 9.7$ V

This voltage appears across each half of the transformer secondary. Across the entire secondary we have $2 \times 9.7 = 19.4$ V (rms).



(b) $V_r = \dfrac{V_p - V_D}{2fCR}$

$2 = \dfrac{13.7 - 0.7}{2 \times 60 \times 200 \times C}$

$\Rightarrow C = \dfrac{12}{2 \times 2 \times 60 \times 200} = 271\ \mu F$

(c) Maximum reverse voltage across D_1 occurs when $v_S = -V_p$. At this point $v_O = V_O$. Thus maximum reverse voltage $= V_O + V_p = 12 + 13.7 = 25.7$. The same applies to D_2.

In specifying the PIV for the diodes, one usually uses a factor of safety of about 1.5,

PIV $= 1.5 \times 25.7 = 38.5$ V

(d) $i_{Dav} = I_L\left[1 + \pi\sqrt{\dfrac{V_p - V_D}{2\ V_r}}\right]$

$= \dfrac{12}{0.2}\left[1 + \pi\sqrt{\dfrac{13.7 - 0.7}{2 \times 2}}\right]$

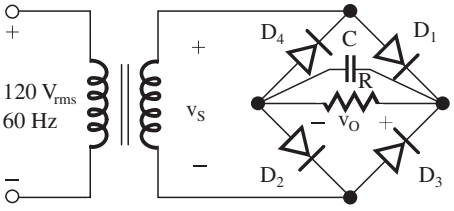
$= 399$ mA

(e) $i_{Dmax} = I_L\left[1 + 2\pi\sqrt{\dfrac{V_p - V_D}{2\ V_r}}\right]$

$= \dfrac{12}{0.2}\left[1 + 2\pi\sqrt{\dfrac{13.7 - 0.7}{2 \times 2}}\right]$

$= 739$ mA

4.80



(a) $V_O = V_p - 2V_D - 1$

$\Rightarrow V_p = V_O + 2V_D + 1 = 12 + 2\times 0.7 + 1 = 14.4$ V

$V_{rms} = \dfrac{14.4}{\sqrt{2}} = 10.2$ V

(b) $V_r = \dfrac{V_p - 2\,V_D}{2fCR}$

$\Rightarrow C = \dfrac{14.4 - 1.4}{2 \times 2 \times 60 \times 200} = 271\ \mu F$

(c) The maximum reverse voltage across D_1 occurs when $V_s = -V_p = -14.4$ V. At this time D_3 is conducting, thus

Maximum reverse voltage $= -V_p + V_{D3}$
$= -14.4 + 0.7 = -13.7$ V

The same applies to the other three diodes. In specifying the PIV rating for the diode we use a factor of safety of 1.5 to obtain

PIV $= 1.5 \times 13.7 = 20.5$ V

(d) $i_{Dav} = I_L\left[1 + \pi\sqrt{\dfrac{V_p - 2\,V_D}{2\,V_r}}\right]$

$= \dfrac{12}{0.2}\left[1 + \pi\sqrt{\dfrac{14.4 - 1.4}{2 \times 2}}\right]$

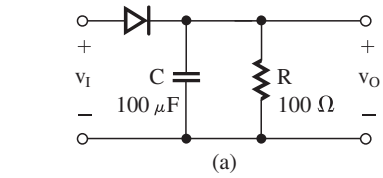
$= 400$ mA

(e) $i_{Dmax} = I_L\left[1 + 2\pi\sqrt{\dfrac{V_p - 2\,V_D}{2\,V_r}}\right]$

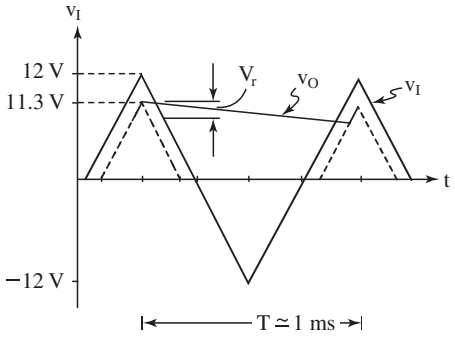
$= \dfrac{12}{0.2}\left[1 + 2\pi\sqrt{\dfrac{14.4 - 0.7}{2 \times 2}}\right]$

$= 740$ mA

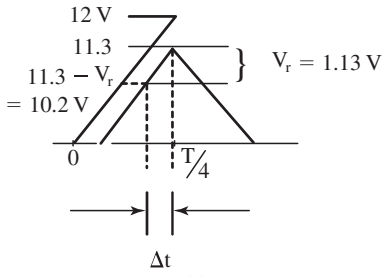
4.81



(a)



(b)



(c)

During the diode s off interval (which is almost equal to T) the capacitor discharges and its voltage is given by

$v_O(t) = 11.3\ e^{-t/CR}$

where $C = 100\ \mu F$ and $R = 100\ \Omega$, thus

$CR = 100 \times 10^{-6} \times 100 = 0.01$ s

At the end of the discharge interval, $t \simeq T$ and

$v_O = 11.3\ e^{-T/CR}$

Since $T = 0.001$ s is much smaller than CR,

$v_O \simeq 11.3\left(1 - \dfrac{T}{CR}\right)$

The ripple voltage V_r can be found as

$V_r = 11.3 - 11.3\left(1 - \dfrac{T}{CR}\right)$

$= \dfrac{11.3T}{CR} = \dfrac{11.3 \times 0.001}{0.01} = 1.13$ V

The average dc output voltage is

$v_O = 11.3 - \dfrac{V_r}{2} = 11.3 - \dfrac{1.13}{2} = 10.74$ V

To obtain the interval during which the diode conducts, $\triangle t$, refer to Fig. (c).

$\dfrac{12}{T/4} = \dfrac{V_r}{\triangle t}$

$\Rightarrow \triangle t = \dfrac{V_r \times (T/4)}{12} = \dfrac{1.13 \times 1}{12 \times 4}$

$= 23.5\ \mu s$

Now, using the fact that the charge gained by the capacitor when the diode is conducting is equal to the charge lost by the capacitor during its discharge interval, we can write

$i_{Cav} \times \triangle t = C\ V_r$

$\Rightarrow i_{Cav} = \dfrac{C\ V_r}{\triangle t} = \dfrac{100 \times 10^{-6} \times 1.13}{23.5 \times 10^{-6}} = 4.8$ A

$i_{Dav} = i_{Cav} + i_{Lav}$

where i_{Lav} is the average current through R during the short interval $\triangle t$. This is approximately $\dfrac{11.3}{R} = \dfrac{11.3}{100} = 0.113$ A. Thus

$i_{Dav} = 4.8 + 0.113 = 4.913$ A

Finally, to obtain the peak diode current, we use

$$i_{Dmax} = i_{Cmax} + i_{Lmax}$$
$$= C\,\frac{dv_I}{dt} + \frac{11.3}{R}$$
$$= C \times \frac{12}{T/4} + \frac{11.3}{R}$$
$$= 100 \times 10^{-6} \times \frac{12 \times 4}{1 \times 10^{-3}} + \frac{11.3}{100}$$
$$= 4.8 + 0.113 = 4.913 \text{ A}$$

which is equal to the average value. This is a result of the linear v_I which gives rise to a constant capacitor current during the diode conduction interval. Thus $i_{Cmax} = i_{Cav} = 4.8$ A. Also, the maximum value of i_L is approximately equal to its average value during the short interval Δt.

4.82 Refer to Fig. P4.74 and let a capacitor C be connected across each of the load resistors R. The two supplies v_O^+ and v_O^- are identical. Each is a full-wave rectifier similar to that of the tapped-transformer circuit. For each supply,

$V_O = 12$ V

$V_r = 1$ V (peak to peak)

Thus

$v_O = 12 \pm 0.5$ V

It follows that the peak value of v_S must be $12.5 + 0.7 = 13.2$ V and the total rms voltage across the secondary will be

$$= \frac{2 \times 13.2}{\sqrt{2}} = 18.7 \text{ V (rms)}$$

$$\text{Transformer turns ratio} = \frac{120}{18.7} = 6.43{:}1$$

To deliver 100-mA dc current to each load,

$$R = \frac{12}{0.1} = 120\ \Omega$$

Now, the value of C can be found from

$$V_r = \frac{V_p - 0.7}{2fCR}$$
$$1 = \frac{12.5}{2 \times 60 \times C \times 120}$$
$$\Rightarrow C = 868\ \mu\text{F}$$

To specify the diodes, we determine i_{Dav} and i_{Dmax},

$$i_{Dav} = I_L(1 + \pi\sqrt{(V_p - 0.7)/2\,V_r}\,)$$
$$= 0.1(1 + \pi\sqrt{12.5/2}\,)$$
$$= 785 \text{ mA}$$

$$i_{Dmax} = I_L(1 + 2\pi\sqrt{(V_p - 0.7)/2\,V_r}\,)$$
$$= 0.1(1 + 2\pi\sqrt{12.5/2}\,)$$
$$= 1.671 \text{ A}$$

To determine the required PIV rating of each diode, we determine the maximum reverse voltage that appears across one of the diodes, say D_1. This occurs when v_S is at its maximum negative value $-V_p$. Since the cathode of D_1 will be at $+12.5$ V, the maximum reverse voltage across D_1 will be $12.5 + 13.2 = 25.7$ V. Using a factor of safety of 1.5, then each of the four diodes must have

$$\text{PIV} = 1.5 \times 25.7 = 38.6 \text{ V}$$

4.83 Refer to Fig. P4.83. When v_I is positive, v_A goes positive, turning on the diode and closing the negative feedback loop around the op amp. The result is that $v_- = v_I$, $v_O = 2v_- = 2v_I$, and $v_A = v_O + 0.7$. Thus

(a) $v_I = +1$ V, $v_- = +1$ V, $v_O = +2$ V, and $v_A = +2.7$ V.

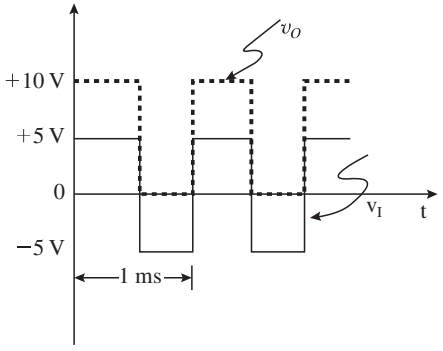
(b) $v_I = +3$ V, $v_- = +3$ V, $v_O = +6$ V, and $v_A = +6.7$ V.

When v_I goes negative, v_A follows, the diode turns off, and the feedback loop is opened. The op amp saturates with $v_A = -13$ V, $v_- = 0$ V and $v_O = 0$ V. Thus

(c) $v_I = -1$ V, $v_- = 0$ V, $v_O = 0$ V, and $v_A = -13$ V.

(d) $v_I = -3$ V, $v_- = 0$ V, $v_O = 0$ V, and $v_A = -13$ V.

Finally, if v_I is a symmetrical square wave of 1-kHz frequency, 5-V amplitude, and zero average, the output will be zero during the negative half cycles of the input and will equal twice the input during the positive half cycles. See figure.

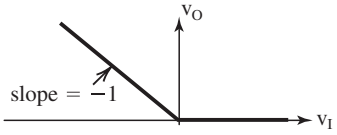


Thus, v_O is a square wave with 0-V and +10-V levels, i.e. 5-V average and, of course, the same frequency (1 kHz) as the input.

4.84 $v_I > 0$: D_1 conducts and D_2 cutoff

$v_I < 0$: D_1 cutoff,

D_2 conducts $\sim \dfrac{v_O}{v_I} = -1$



(a) $v_I = +1$ V

$v_O = 0$ V

$v_A = -0.7$ V

Keeps D_2 off so no current "ows through R

$\Rightarrow v_- = 0$ V

Virtual ground as feedback loop is closed through D_1

(b) $v_I = +3$ V

$v_O = 0$ V

$v_A = -0.7$ V

$v_- = 0$ V

(c) $v_I = -1$ V

$v_O = +1$ V

$v_A = 1.7$ V

$v_- = 0$ V

$\sim$ Virtual ground as negative feedback loop is closed through D_2 and R.

(d) $v_I = -3$ V $\Rightarrow v_O = +3$ V

$v_A = +3.7$ V

$v_- = 0$ V

4.85 The input is an AM signal with carrier frequency $\omega_0 = 2\pi \times 10^6$ rad/s and time-varying amplitude $V_p(t) = 1.5 + \sin(2\pi \times 10^3 t)$ Volts. With $CR = 0.1$ ms, the output time constant is much longer than the period of the carrier, $1/10^6 = 1$s. Thus, the output drops very little between successive peaks of the carrier. However, the time constant is much smaller than the period of $V_p(t)$ so that v_O will follow changes in $V_p(t)$. A sketch appears in Fig. 1.

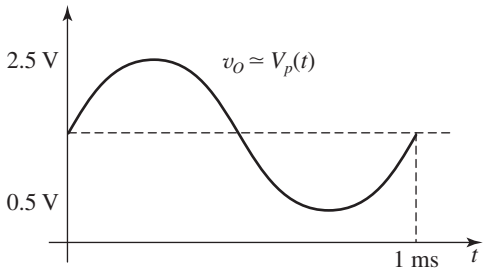


Figure 1

With $CR = 0.1\,\mu$s, the output time constant is much smaller than the period of the carrier (1 μs). Thus, the circuit behaves as a precision rectifier. A sketch appears in Fig. 2.

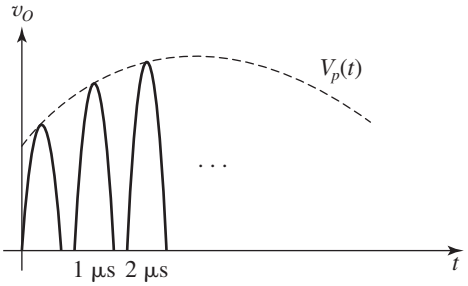
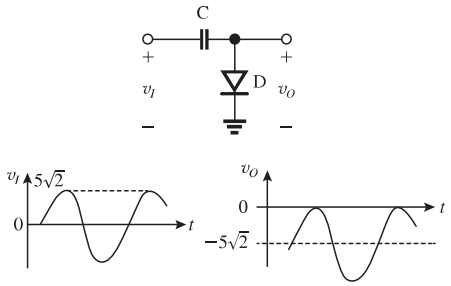


Figure 2

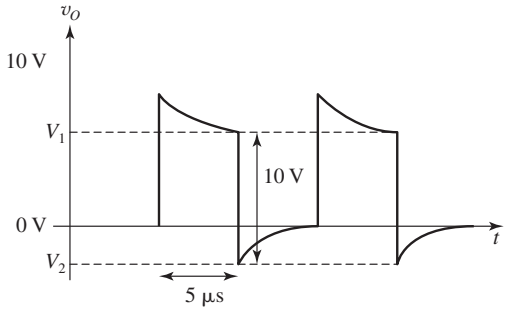
4.86



From the figure we see that

$v_{Oav} = -5\sqrt{2} = -7.07$ V

4.87

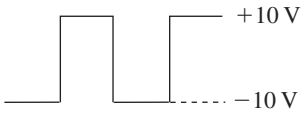


$V_1 = 5 + 5e^{-5/10} = 8.03$ V

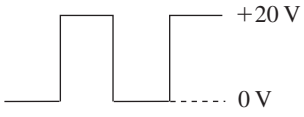
$V_2 = V_1 - 10 = -1.97$ V

4.88

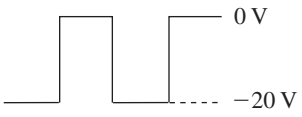
(a)



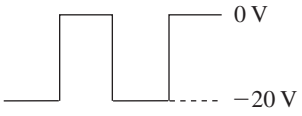
(b)



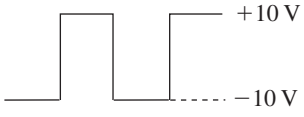
(c)



(d)



(e)



(f) Here there are two different time constants involved. To calculate the output levels, we shall consider the discharge and charge wave forms.

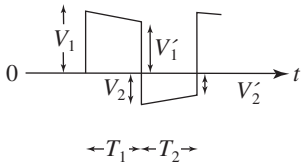
During T_1, $\quad v_O = V_1 e^{-t/RC}$

At $t = T_1 = T, v_O = V_1'$

$= V_1 e^{-T/RC}$

where for $T \ll CR$

$V_1' \simeq V_1(1 - T/CR) = V_1(1 - \alpha)$



where $\alpha \ll 1$

During the interval T_2, we have

$|v_O| = |V_2|\, e^{-t/(CR/2)}$

At the end of $T_2, t = T,$ and $v_O = |V_2'|$

where

$|V_2'| = |V_2|\, e^{-T/(CR/2)}$

$$\simeq |V_2| \left(1 - \frac{T}{RC/2}\right) = |V_2|\,(1 - 2\alpha)$$

Now

$V_1' + |V_2| = 20 \Rightarrow V_1 + |V_2| - \alpha V_1 = 20 \qquad (1)$

and

$\left|V_2'\right| + V_1 = 20 \Rightarrow V_1 + |V_2| - 2\alpha\,|V_2| = 20 \quad (2)$

From (1) and (2) we find that

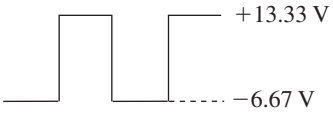
$V_1 = 2|V_2|$

Then using (1) and neglecting αV_1 yields

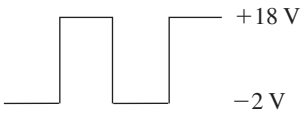
$3\,|V_2| = 20 \Rightarrow\ |V_2| = 6.67$ V

$V_1 = 13.33$ V

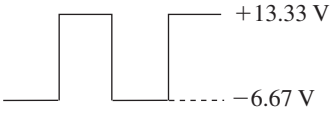
The result is



(g)



(h) Using a method similar to that employed for case (f) above, we obtain



4.89 $C_{j0} = 150$ fF, $V_0 = 3$ V , $m = 3$. Using Equation (1.71),

$$C_j = \frac{150 \text{ fF}}{\left(1 + \frac{V_R}{3}\right)^3}$$

$$\Rightarrow V_R = 3\left(\frac{150}{C_j}\right)^{1/3} - 3$$

at $C_j = 80$ fF: $V_R = 3\left(\frac{150}{80}\right)^{1/3} - 3 = 0.70$ V,

and

at $C_j = 20$ fF: $V_R = 3\left(\frac{150}{20}\right)^{1/3} - 3 = 2.87$ V.

Thus, the range of required bias voltages is $0.70 \text{ V} < V_R < 2.87$ V.

4.90 (a) Assuming an ideal op amp, there is a virtual short-circuit between input terminals so that the photodiode reverse voltage is,

$$V_R = v_- = v_+ = 3 \text{ V}$$

Thus,

$$v_O = 3 - i_D \times 250 = 3 + (I_D + i_P) \times 250 = 3 + (I_D + R \times P) \times 250$$

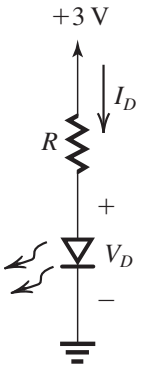
(b) With no illumination, $P = 0$ and

$$v_O = 3 + I_D \times 250 = 3 + 10^{-7} \times 250 \cong 3 \text{ V}$$

(c) With $P = 10$ W,

$$v_O = 3 + (10^{-7} + 10^{-2} \times 0.5) \times 250 = 4.25 \text{ V}$$

4.91



(a) $V_D = 2.2$ V, $I_D = 10$ mA.

$$\Rightarrow R = \frac{3 - 2.2}{10} = 80\Omega$$

(b) $V_D = 1.8$ V, $I_D = 10$ mA.

$$\Rightarrow R = \frac{3 - 1.8}{10} = 120\Omega$$